


# UNIVERSITI TEKNIKAL MALAYSIA MELAKA

## FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

# W O R K S H O P 1

# R E P O R T

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**Program** : BITS S2G2

**Project Title** : BLOOD BANK MANAGEMENT SYSTEM

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# EXECUTIVE SUMMARY

In today’s world, the demand for blood is rising significantly due to various illnesses and diseases like cancer. Blood donation saves lives, but managing donation records has become increasingly difficult. Many organizations still use manual record-keeping, where staff write down donor and donation details on paper. This traditional method often leads to problems such as illegible handwriting, duplicated data, lost records over time, and mistakes from manual entry. As the amount of data increases, managing and retrieving information becomes slow and inefficient.

To solve these issues, this project suggests developing a Blood Bank Management System that offers a computerized platform for managing and storing donor and blood donation records efficiently. The system aims to make tasks easier for staff and nurses by providing a user-friendly interface for accurate and secure data handling. It also improves communication between staff and donors, reduces data duplication, and enhances overall data management.

The system is built using C++ as the console application and MySQL as the database, ensuring fast performance and reliable storage. It includes modules for managing important entities like donors, recipients, doctors, and donation locations. The main users are staff members and donors, who can easily enter, update, and retrieve data.

The project's significance lies in its ability to digitalize the entire blood donation record process, improving accuracy, accessibility, and efficiency. By implementing this system, hospitals and blood donation centres can manage their resources better, reduce errors, and provide timely support for patients needing blood.

# TABLE OF CONTENTS

**PAGE**

**EXECUTIVE SUMMARY ..... i**

**TABLE OF CONTENTS ..... ii**

**LIST OF TABLES ..... vii**

**LIST OF FIGURES ..... viii**

**CHAPTER 1: INTRODUCTION ..... 1**

1.1 Introduction ..... 1

1.2 Problem Statement ..... 2

1.3 Objective (s) of the project ..... 3

1.4 Scope ..... 4

1.4.1 Modules to be developed ..... 4

1.4.2 Target User ..... 5

1.5 Project Significance ..... 6

1.6 Gantt Chart of Project Activities ..... 6

**2 ANALYSIS OF PROBLEM ..... 7**

2.4 Problem Decomposition Description ..... 7

2.5 Structured Chart ..... 8

**3 DESIGN ..... 9**

3.1 Business Rules ..... 9

3.2 Flowchart ..... 12

3.3 ERD ..... 17

3.4 Data Normalization ..... 17

3.5 Data Dictionary ..... 20

3.6 Interface Design ..... 22

3.6.1 Main Page ..... 22

3.6.2 Login Menu Page ..... 22

3.6.3 Admin Access Page ..... 23

3.6.4 Admin Main Menu ..... 23

3.6.5 Admin Control Menu ..... 24

3.6.6 Admin Add Staff ..... 24

3.6.7 Admin Delete Staff ..... 25

3.6.8 Admin Update Staff ..... 26

3.6.9 Admin Search Staff ..... 26

3.6.10 Admin Display Staff Record ..... 27

3.6.11 Admin Generate Report ..... 27

3.6.12 Admin Generate Report By Gender ..... 27

3.6.13 Admin Generate Report By Position ..... 28

3.6.14 Admin Generate Report By Age ..... 28

3.6.15 Staff Access Page ..... 29

3.6.16 Staff Main Menu ..... 29

3.6.17 Staff Control Menu ..... 30

3.6.18 Staff Add Record Menu ..... 30

3.6.19 Staff Add Record Menu-Add Donor ..... 31

3.6.20 Staff Add Record Menu-Add Block Stock ..... 31

3.6.21 Staff Add Record Menu-Add Donation ..... 32

3.6.22 Staff Delete Record Menu ..... 32

3.6.23 Staff Delete Record Menu-Delete Donor Record ..... 33

3.6.24 Staff Delete Record Menu-Delete Blood Stock Record ..... 34

3.6.25 Staff Delete Record Menu-Delete Donation Record ..... 35

3.6.26 Staff Update Record Menu ..... 36

3.6.27 Staff Update Record Menu-Update Donor ..... 36

3.6.28 Staff Update Record Menu-Update Blood Stock ..... 37

3.6.29 Staff Update Record Menu-Update Donation ..... 38

3.6.30 Staff Search Record Menu ..... 38

3.6.31 Staff Search Record Menu-Staff Search Donor Record ..... 39

3.6.32 Staff Search Record Menu-Staff Search Blood Stock Record ..... 39

3.6.33 Staff Search Record Menu-Staff Search Donation Record ..... 40

3.6.34 Blood Donation Report Page ..... 40

3.6.35 Staff Account Information Page ..... 41

3.6.36 Donor Access Page ..... 41

3.6.37 Donor Main Menu ..... 42

3.6.38 Donor Recent Donation Record Page ..... 42

3.6.39 Donor Account Information Page ..... 42

3.6.40 Donor Reward Points ..... 43

3.6.41 Account Registration Menu Page ..... 43

3.6.42 New Admin/Staff Registration ..... 44

3.6.43 New Donor Registration ..... 44

3.6.44 Password Recovery Menu Page ..... 45

3.6.45 Admin/Staff Password Recovery ..... 45

3.6.46 Donor Recovery ..... 45

**4 IMPLEMENTATION ..... 46**

4.6 Naming Convention ..... 46

4.6.1 Pascal Case ..... 46

4.7 Function ..... 47

4.8 Selection ..... 47

4.8.1 If/Else Technique ..... 48

4.8.2 Switch Technique ..... 48

4.9 Control ..... 49

4.10 Error Handling ..... 49

4.11 Business Rule Implementation ..... 50

4.12 Complex Calculation Implementation ..... 51

4.13 Report Generation for Analysis Implementation ..... 52

**5 CONCLUSION ..... 53**

5.6 Constraints ..... 53

5.7 Future Improvements ..... 55

**REFERENCES ..... 57**

# LIST OF TABLES

**PAGE**

Table 1.1: Gantt Chart of Project Activities ..... 6

Table 3.1: Authentication and Validation Module Business Rule Table..... 9

Table 3.2: User Profile Management Module Business Rule Table..... 10

Table 3.3: Reward Point Module Business Rule Table ..... 10

Table 3.4: Reward Point Module Business Rule Table ..... 11

Table 3.5: Report Generation Module Business Rule Table ..... 11

# LIST OF FIGURES

**PAGE**

Figure 2.1: Structured Chart ..... 8

Figure 3.1: Admin Main Menu ..... 12

Figure 3.2: Admin Control Menu ..... 13

Figure 3.3: Staff Main Menu ..... 14

Figure 3.4: Staff Control Menu ..... 15

Figure 3.5: Donor Main Menu ..... 16

Figure 3.6: Entity Relationship Diagram ..... 17

Figure 3.7: Data Dictionary for Table Staff ..... 20

Figure 3.8: Data Dictionary for Table Donor ..... 20

Figure 3.9: Data Dictionary for Table BloodStock ..... 21

Figure 3.10: Data Dictionary for Table Donation ..... 21

Figure 3.11: Data Dictionary for Table RewardPoints ..... 21

Figure 3.12: Main Page Interface ..... 22

Figure 3.13: Login Menu Interface ..... 22

Figure 3.14: Admin Access Interface ..... 23

Figure 3.15: Admin Main Menu Interface ..... 23

Figure 3.16: Admin Control Interface ..... 24

Figure 3.17: Admin Add Staff ..... 24

Figure 3.18: Admin Delete Staff ..... 25

Figure 3.19: Admin Update Staff ..... 26

Figure 3.20: Admin Search Staff ..... 26

Figure 3.21: Admin Display Staff Record Interface ..... 27

Figure 3.22: Admin Generate Report Interface ..... 27

Figure 3.23: Admin Generate Report By Gender ..... 27

Figure 3.24: Admin Generate Report By Position ..... 28

Figure 3.25: Admin Generate Report By Age ..... 28

Figure 3.26: Staff Access Interface ..... 29

Figure 3.27: Staff Main Menu Interface ..... 29

Figure 3.28: Staff Control Menu Interface ..... 30

Figure 3.29: Staff Add Record Menu Interface ..... 30

Figure 3.30: Staff Add Record Menu-Add Donor ..... 31

Figure 3.31: Staff Add Record Menu-Add Block Stock ..... 31

Figure 3.32: Staff Add Record Menu-Add Donation ..... 32

Figure 3.33: Staff Delete Record Menu Interface ..... 32

Figure 3.34: Staff Delete Record Menu-Delete Donor Record ..... 33

Figure 3.35: Staff Delete Record Menu-Delete Blood Stock Record ..... 34

Figure 3.36: Staff Delete Record Menu-Delete Donation Record.....35

Figure 3.37: Staff Update Record Menu Interface.....36

Figure 3.38: Staff Update Record Menu-Update Donor.....36

Figure 3.39: Staff Update Record Menu-Update Blood Stock.....37

Figure 3.40: Staff Update Record Menu-Update Donation.....38

Figure 3.41: Staff Search Record Menu Interface.....38

Figure 3.42: Staff Search Record Menu-Staff Search Donor Record.....39

Figure 3.43: Staff Search Record Menu-Staff Search Blood Stock Record.....39

Figure 3.44: Staff Search Record Menu-Staff Search Donation Record.....40

Figure 3.45: Blood Donation Report Interface.....40

Figure 3.46: Staff Account Information Interface.....41

Figure 3.47: Donor Access Interface.....41

Figure 3.48: Donor Main Menu.....42

Figure 3.49: Donor Recent Donation Record Interface.....42

Figure 3.50: Donor Account Information Interface.....42

Figure 3.51: Donor Reward Points Interface.....43

Figure 3.52: Account Registration Menu Interface.....43

Figure 3.53: New Admin/Staff Registration.....44

Figure 3.54: New Donor Registration.....44

Figure 3.55: Password Recovery Menu Interface.....45

Figure 3.56: Admin/Staff Password Recovery.....45

Figure 3.57: Donor Recovery .....45

Figure 4.1: Pascal Case Implementation.....46

Figure 4.2: Function Implementation .....47

Figure 4.3: If/Else Technique Implementation .....48

Figure 4.4: Switch Technique Implementation.....48

Figure 4.5: Control Implementation .....49

Figure 4.6: Error Handling Implementation .....49

Figure 4.7: Complex Calculation Implementation.....51

Figure 4.8: Report Generation for Analysis Implementation .....52

# CHAPTER 1: INTRODUCTION

## 1.1 Introduction

Donating blood is a vital act that can save many lives. It involves two main parties: donors and recipients. Their compatibility is based on the antigens on the surface of red blood cells. It plays an important role in healthcare systems around the world especially for patients who need transfusions due to surgery, accidents, severe illness, or conditions like anemia and cancer. It ensures that hospitals have a steady and safe supply of blood for emergencies and planned treatments. Today, thousands of people need donated blood to stay healthy. However, the medical sector in Malaysia currently faces difficulties in keeping track of blood donors' personal information during blood donation campaigns. This issue arises because many organizations still use traditional, paper-based systems. The manual method of recording information on paper often results in unreadable handwriting, messy records, and difficulties maintaining data over time. This makes it hard for both staff and donors to access correct information. Additionally, manual record-keeping is insecure and takes a lot of time. Staff members are required to search through numerous records just to find a donor's details, which is necessary to ensure that the person is eligible to donate blood. Since an individual can only donate once every six months, verifying this information manually can be a slow and tedious process. As a result, staff spend a great deal of time handling paperwork and cross-checking files, which reduces efficiency and increases the chances of error. Collecting, storing, and distributing blood requires careful management to ensure safety and efficiency. Without a good system, tracking donor information, blood group availability, and expiry dates becomes hard. This can lead to shortages or waste of valuable blood units.

So, creating a simple and reliable Blood Bank Management System is essential to support blood donation. It helps keep accurate records, improves access to information, and makes sure that the right type of blood is available when needed. This project aims to use technology to strengthen the blood donation process and improve the effectiveness of blood bank operations. This system helps staff and donors manage donation details easily. It uses Oracle SQL as a centralized database for the secure storage and retrieval of donor and staff records. Since many people use mobile devices, the system also allows donors to report issues. The interface, created with C++, promotes effective communication between donors and staff.

## 1.2 Problem Statement

i. **Difficulty in retrieving information** related to blood donation between staff and donor causes wastage of time and inefficiency. Staff need to manually search through physical past records one-by-one, and it could be hard especially if donation records has been updated.

ii. **Difficulty in generating and analyzing reports.** Since all records are kept manually, preparing reports on blood donation activities, donor frequency, or blood stock levels takes a lot of time and is prone to human error.

iii. **Inefficiency in documentation.** Staff found it hard to record information related to donors, staff, donation and locations. All donation details are recorded manually by handwriting, but handwriting can fade over a period of time and can be hard to read. As a result, documentation of records can be even more tedious to be completed and can consume a lot of time.

iv. **Risk of data loss** due to poor organization and insecure storage. Unorganized documents and paperwork can result in losing important information and might bring potential harm to donors and staff.

## 1.3 Objective (s) of the project

This project embarks on the following objectives:

1. To handle donors and staff details efficiently. The system allows staff to easily search and update donor records using attributes like name, address, and donation date. Hence, staff can quickly filter and find the correct record. This helps to reduce the time spent on manual searches. Donors can also view and receive notifications about updates to their donation details, which improves transparency and engagement in blood donation.

2. To generate reports based on user preferences. The system can make reports on donor information, blood donation records and more. Staff can analyze data efficiently as these reports can be filtered by specific attributes. Reports are also stored in the database to prevent information loss and to help staff in managing performance metrics like total number of donors or number of donations monthly or yearly.

3. To improve data accuracy and reduce human error. By replacing manual record-keeping with an automated system, the risk of mistakes in donor details, donation dates, and eligibility status is greatly lowered. This ensures reliable and consistent information management.

4. To boost operational efficiency and decision-making. The system simplifies administrative tasks, allowing staff to concentrate on donor management and campaign coordination. Accurate data and generated reports also help with better planning and informed choices for future blood donation drives.

5. To maintain all records in a centralized database. The system stores donor, staff, donation, and location details in one easy to access database. Hence, staff can track donation histories and check donor eligibility from time to time, like whether six months have passed since the donor's last donation. This approach enhances accuracy, consistency and secure data management.

6. To introduce a donor reward point system that encourages regular blood donations. The system gives points to donors every time they donate blood or take part in donation campaigns. Donors can accumulate these points and later exchange them for recognition or small rewards. This motivates donors to give blood more often and stay involved with the blood bank.

## 1.4 Scope

### 1.4.1 Modules to be developed

**Authentication and Validation Modules:**

- Login and Logout
- User registration
- Forgot password

**User Profile (Staff/Admin and Donor) Management:**

- Add blood donation record
- Display blood donation record
- Delete blood donation record
- Update blood donation record
- Search blood donation record
- Manage donor reward point system

**Report Generation for Records:**

- Generate report on staff management and blood donation history

### 1.4.2 Target User

**Staffs/Admin**

- Login and logout
- View staffs' details
- Create, update, delete, and retrieve blood donation records.
- Create, update, delete, and query donor records to manage donor information.
- Create, update, delete, and query staff's records to manage staff information.
- Create, update, delete, query and retrieve blood stock records to manage blood information.
- Generate reports based on staff management and blood donation history.

**Donors**

- View the donors' donation history.
- View and update donors' own personal information.
- View the report for donors' own donation details.
- View donors' total points and reward history.

## 1.5 Project Significance

1. The system supports paperless operations and reduces the need for physical storage space by replacing manual paperwork with a digital system.

2. The system allows staff to search and view the detailed records of blood donation for each donor and recipients in an organised report format.

3. The staff can easily update and delete donor and staff information to ensure data accuracy.

4. The system enhances the efficiency of managing blood donation by reducing the cost and time required to collect and organize information from donors and staff.

5. The system improves data security by storing staff and donors information in a secure database, hence preventing data loss and unauthorized access.

6. Donors can easily get access to their blood donation details such as donation history, receive updates and feel more connected with their contributions .

7. The system introduces a reward point system that encourages regular donations by recognizing and rewarding consistent donors to increase donor retention, promotes public awareness, and helps maintain a steady blood supply.

8. The system is able to generate statistical reports on blood stocks to help manage inventory efficiently.

## 1.6 Gantt Chart of Project Activities

**Table 1.1: Gantt Chart of Project Activities**

| Task | week 1 | week 2 | week 3 | week 4 | week 5 | week 6 | week 7 | week 8 | week 9 | week 10 | week 11 | week 12 | week 13 | week 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Briefing of Workshop | | | | | | | | | | | | | | |
| Discussion of project proposal with supervisor | | | | | | | | | | | | | | |
| Proposal Preparation | | | | | | | | | | | | | | |
| Submission Proposal | | | | | | | | | | | | | | |
| Analysis and Design | | | | | | | | | | | | | | |
| Project Implementation 1 | | | | | | | | | | | | | | |
| Project Implementation 2 | | | | | | | | | | | | | | |
| Project Implementation 3 | | | | | | | | | | | | | | |
| Final Presentation &Report Submission | | | | | | | | | | | | | | |

# 2 ANALYSIS OF PROBLEM

## 2.4 Problem Decomposition Description

In the current blood donation management process, managing donor and staff information takes a lot of time and isn't efficient. Medical staff often struggle to update, delete, or retrieve donor records because they have to go through physical files one by one. This increases their workload and raises the chance of human mistakes like incorrect data entry or misplaced documents, which can result in wrong donor and donation records. Donors also cannot easily access their own donation history or eligibility status unless they visit the blood donation centre in person. Additionally, handwritten records can fade or get damaged over time, leading to data loss and making it even harder to keep accurate records. Therefore, a digital Blood Bank Management System is needed to simplify these processes, improve accuracy, security, and accessibility.

1. Database Management: Focuses on centralizing donor, staff, hospital, and blood donation data into one database. This allows for secure storage, easy retrieval, and efficient updates of all records.

2. Report Generation: Provides summarized and detailed reports of blood donations, donor activities, and hospital requests. This helps staff monitor performance and keep accurate records.

3. Search and Filtering Module: Lets staff search for donor, hospital, or donation details using specific attributes such as donor name, blood type, or donation date for quicker data access.

4. Authentication and Authorization: Limits system access to authorized personnel only. This ensures that sensitive donor and staff information is protected from unauthorized use.

5. Reward Point Management: Automatically tracks and updates donor reward points after each donation. This encourages consistent participation and strengthens donor engagement.

## 2.5 Structured Chart

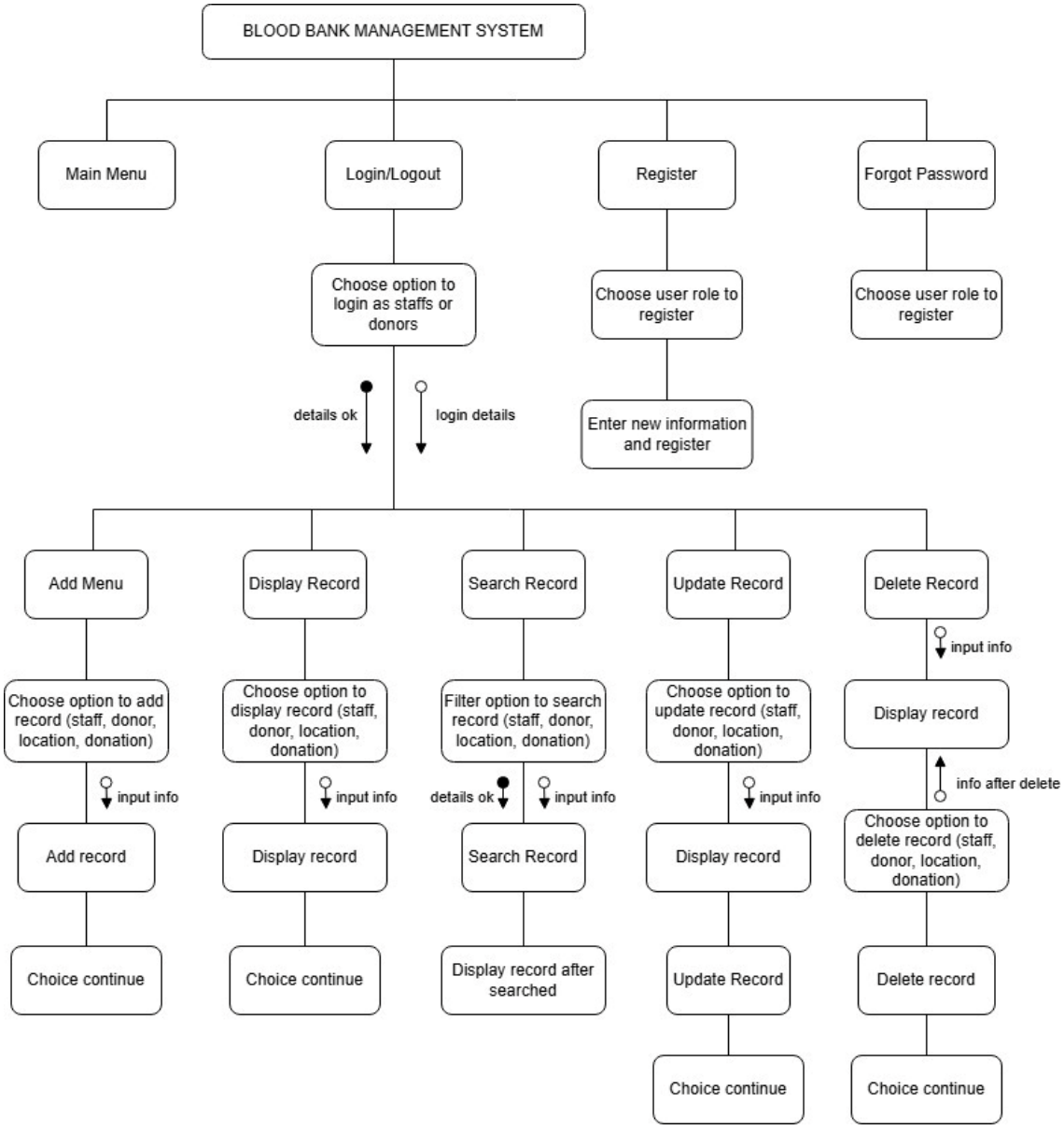



**Figure 2.1: Structured Chart**

# 3 DESIGN

## 3.1 Business Rules

**Authentication and Validation Module**

**Table 3.1: Authentication and Validation Module Business Rule Table**

| Business Rule ID | Business Rule Description |
|---|---|
| BR-AV1 | Only registered users (admin/staff/donor) are allowed to log in to the system. |
| BR-AV2 | Password must match an existing user account before access is granted. |
| BR-AV3 | Passwords must be securely stored and not displayed in plain text. |
| BR-AV4 | Each username must be unique across the system. |
| BR-AV5 | Users must reset their password through identity verification before a new password can be created. |

**User Profile Management Module**

**Table 3.2: User Profile Management Module Business Rule Table**

| Business Rule ID | Business Rule Description |
|---|---|
| BR-UM1 | Only authorized staff/admin are allowed to add, update, search or delete blood donation records. |
| BR-UM2 | System-generated user IDs cannot be duplicated or altered. |
| BR-UM3 | All input fields must be validated to prevent incorrect or duplicate data. |
| BR-UM4 | Users must be able to change their password after proper verification. |

**Blood Donation and Inventory Management Module**

**Table 3.3: Reward Point Module Business Rule Table**

| Business Rule ID | Business Rule Description |
|---|---|
| BR-DI1 | A blood stock can only be added to a blood donation, and each blood donation must be linked to existing blood stock. |
| BR-DI2 | A donor can have multiple donation records, but each blood donation includes only one donor. |
| BR-DI3 | A staff manage one or more blood stocks, while each blood stock is managed by only one staff. |

**Reward Point Module**

**Table 3.4: Reward Point Module Business Rule Table**

| Business Rule ID | Business Rule Description |
|---|---|
| BR-RP1 | Reward points may only be awarded after a valid and completed blood donation record. |
| BR-RP2 | Only donors are allowed to earn reward points, staff/admin cannot. |
| BR-RP3 | Reward points and reward level must be updated based on cumulative total points earned by donor. |

**Report Generation Module**

**Table 3.5: Report Generation Module Business Rule Table**

| Business Rule ID | Business Rule Description |
|---|---|
| BR-RG1 | Only Admin/Staff can generate official reports for donation history and staff management. |
| BR-RG2 | Reports must display complete, accurate and up-to-date data retrieved from the system database. |

## 3.2 Flowchart

Admin Main Menu

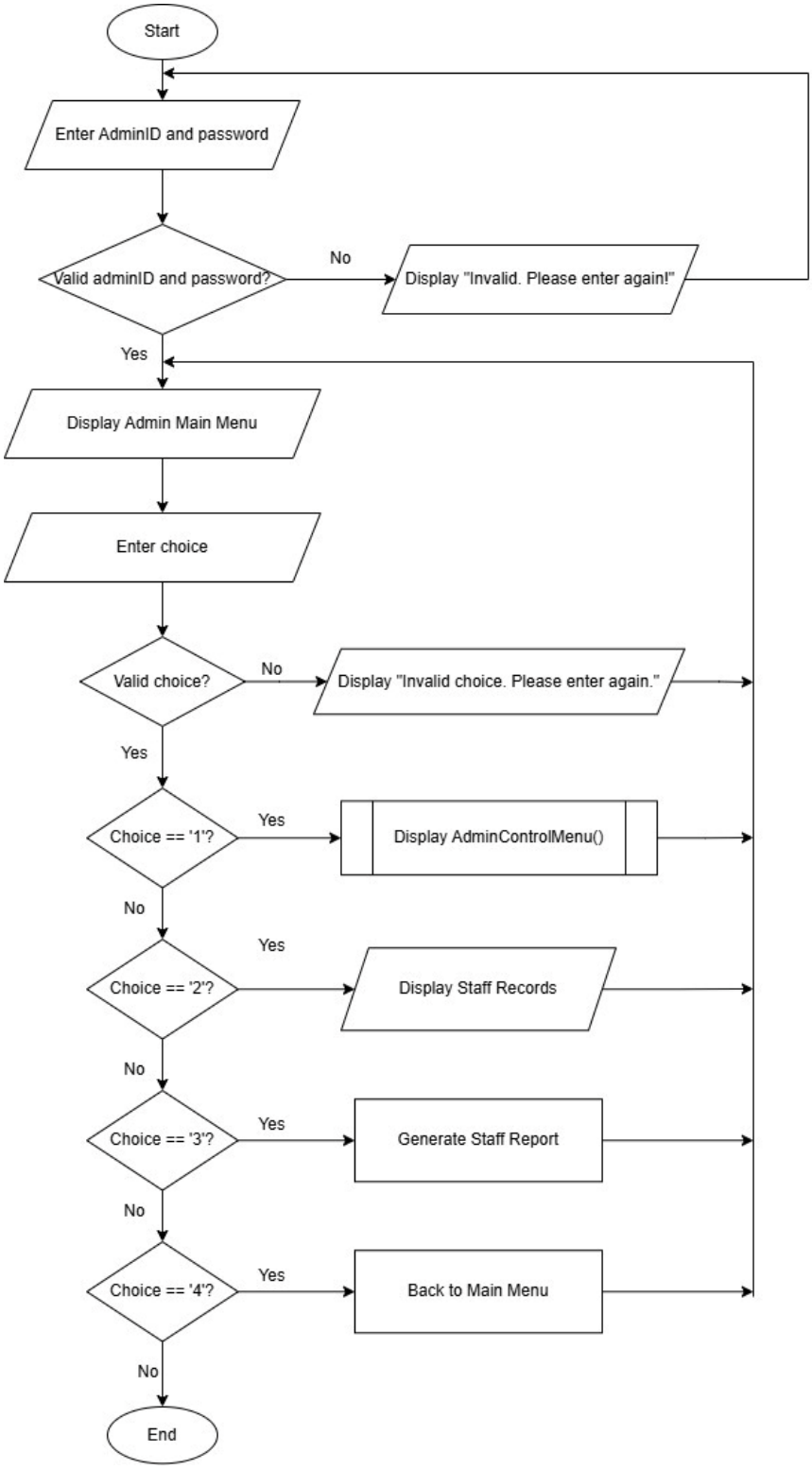



**Figure 3.1: Admin Main Menu**

Admin Control Menu

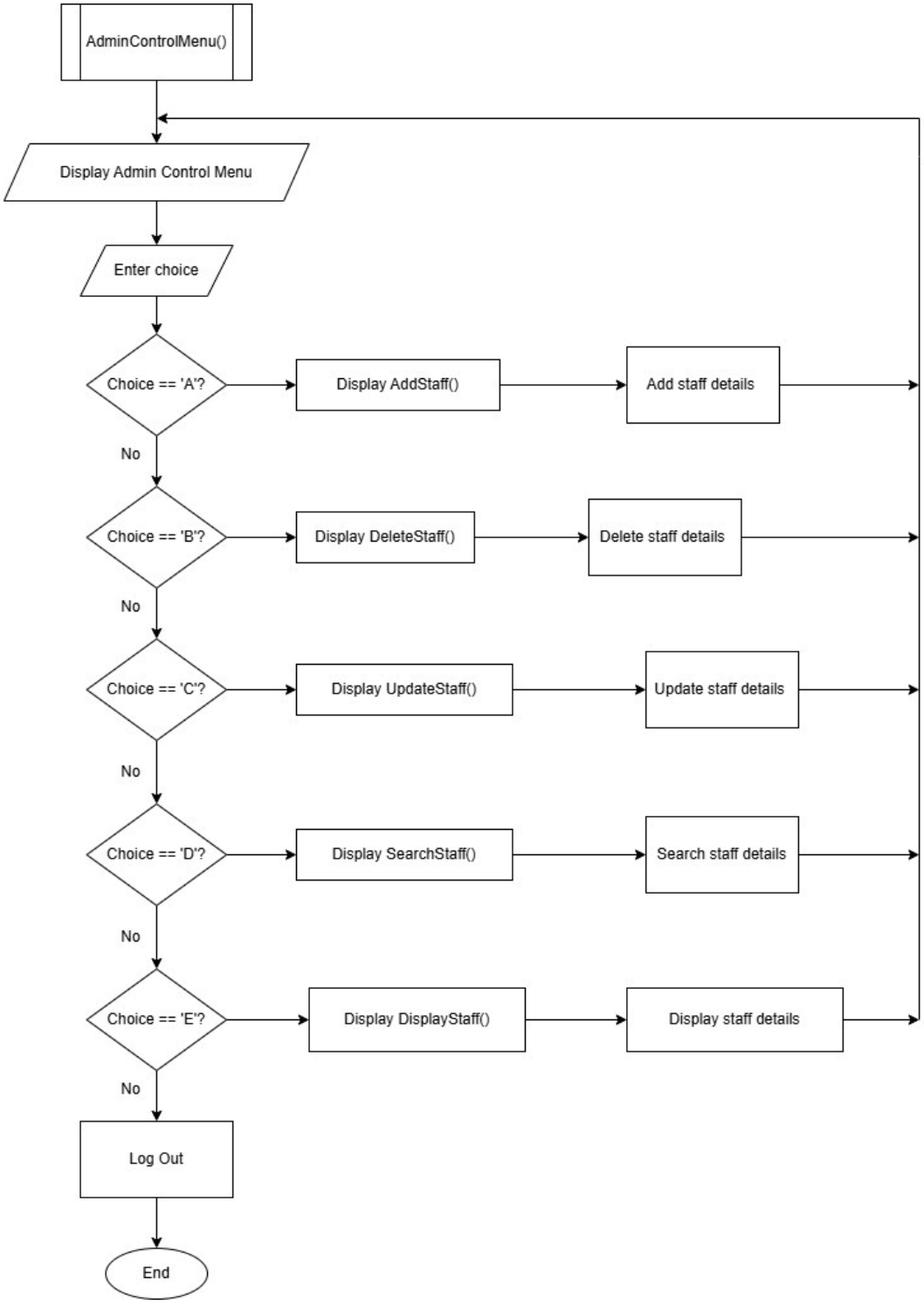



**Figure 3.2: Admin Control Menu**

Staff Main Menu

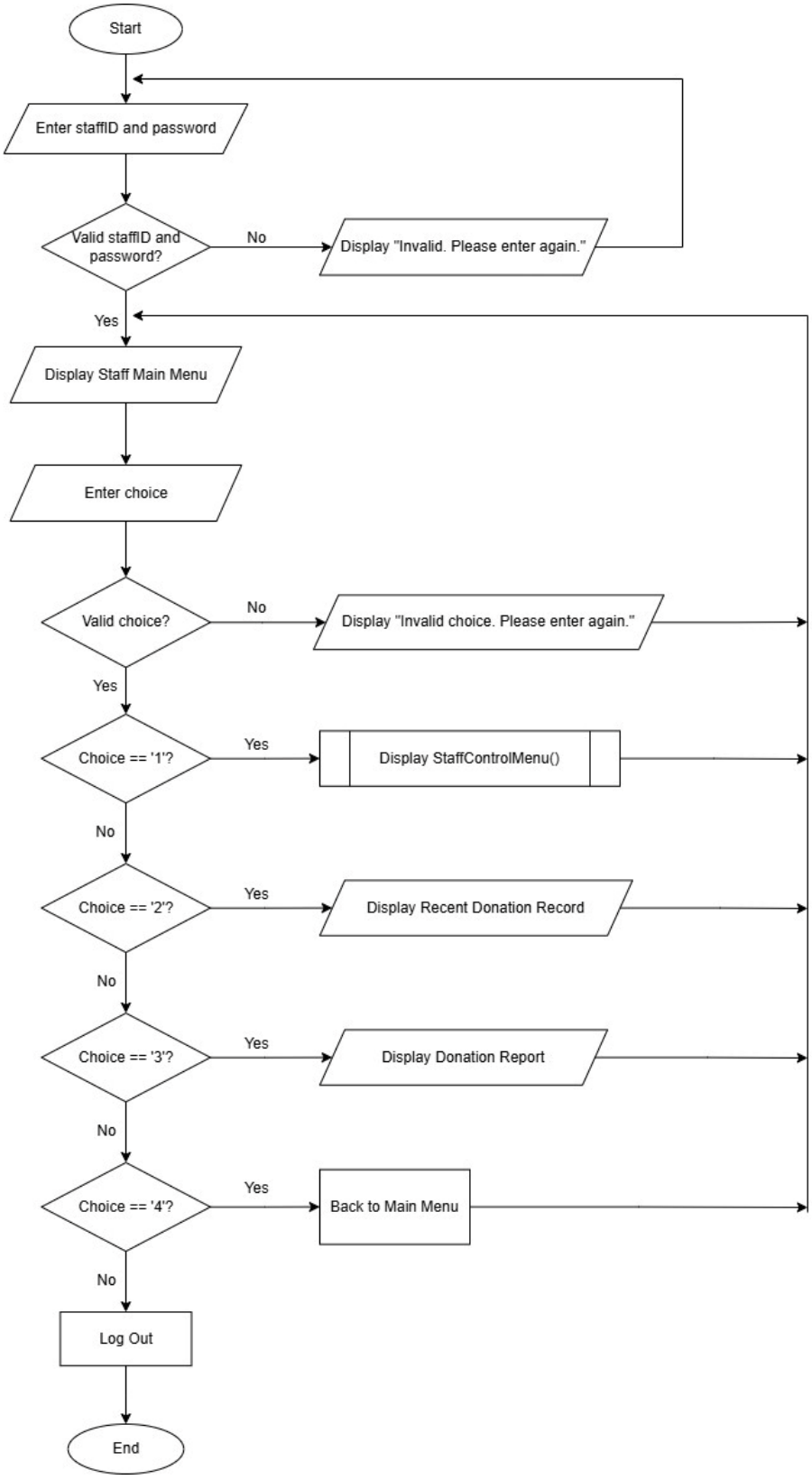



**Figure 3.3: Staff Main Menu**

Staff Control Menu

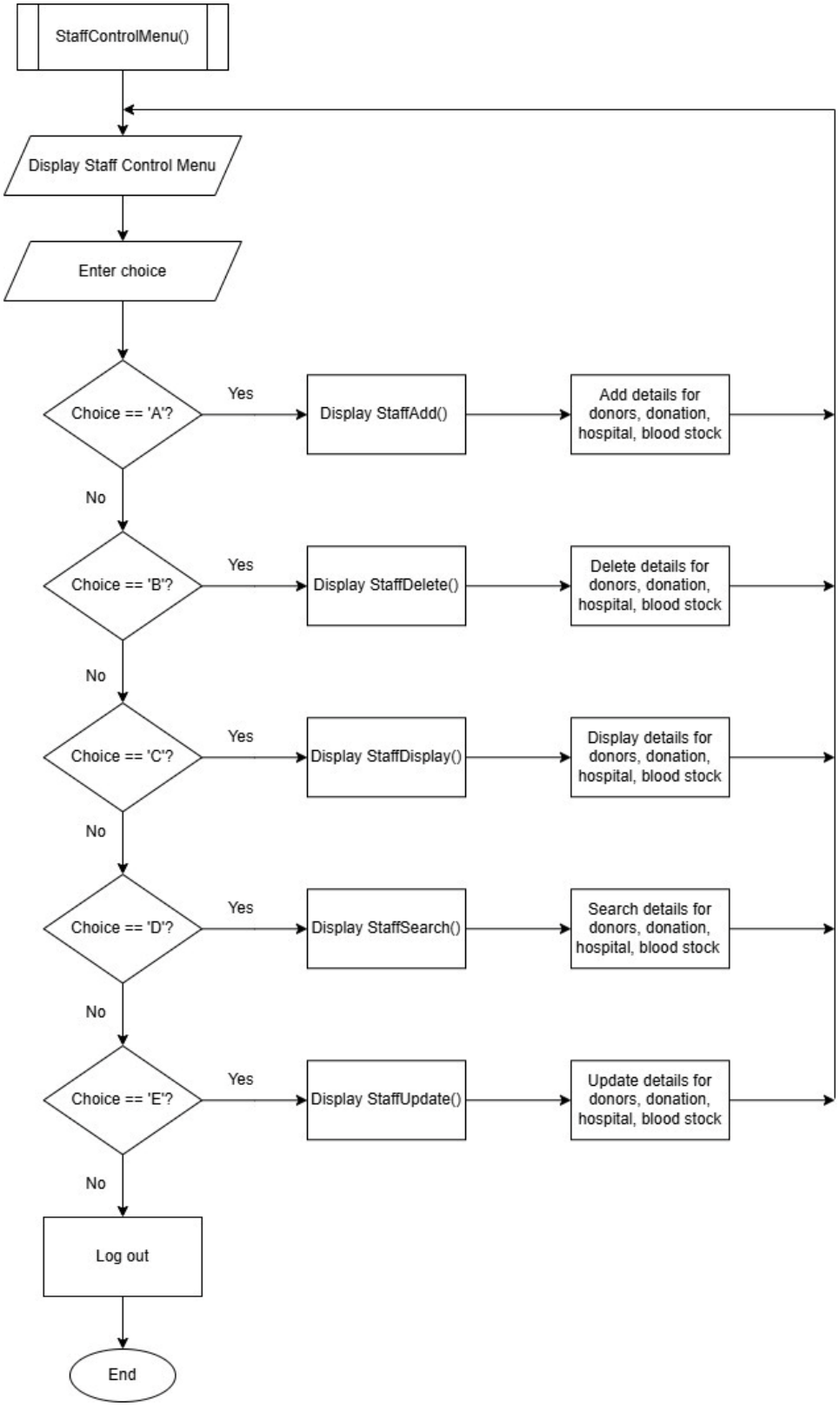



**Figure 3.4: Staff Control Menu**

Donor Main Menu

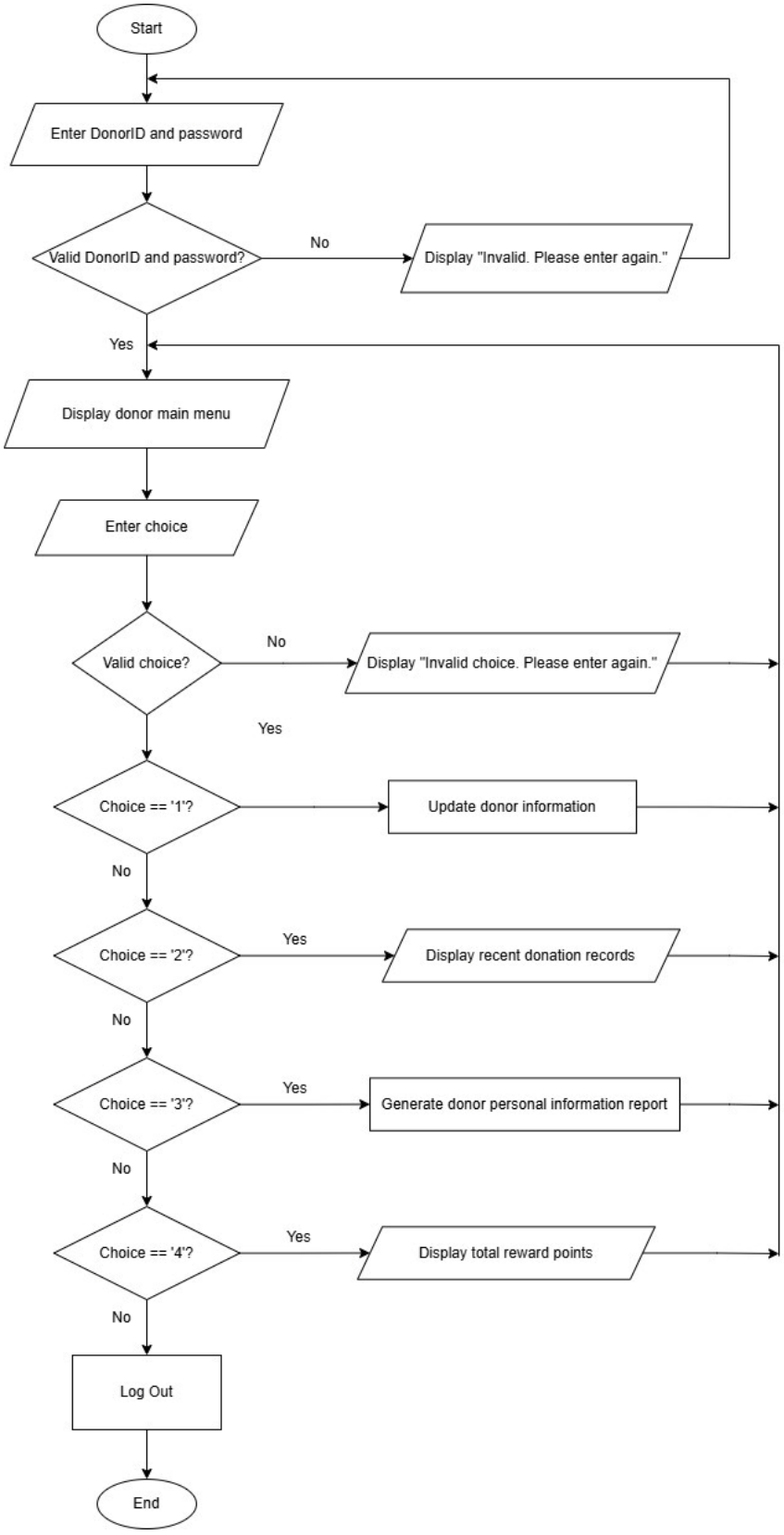



**Figure 3.5: Donor Main Menu**

## 3.3 ERD

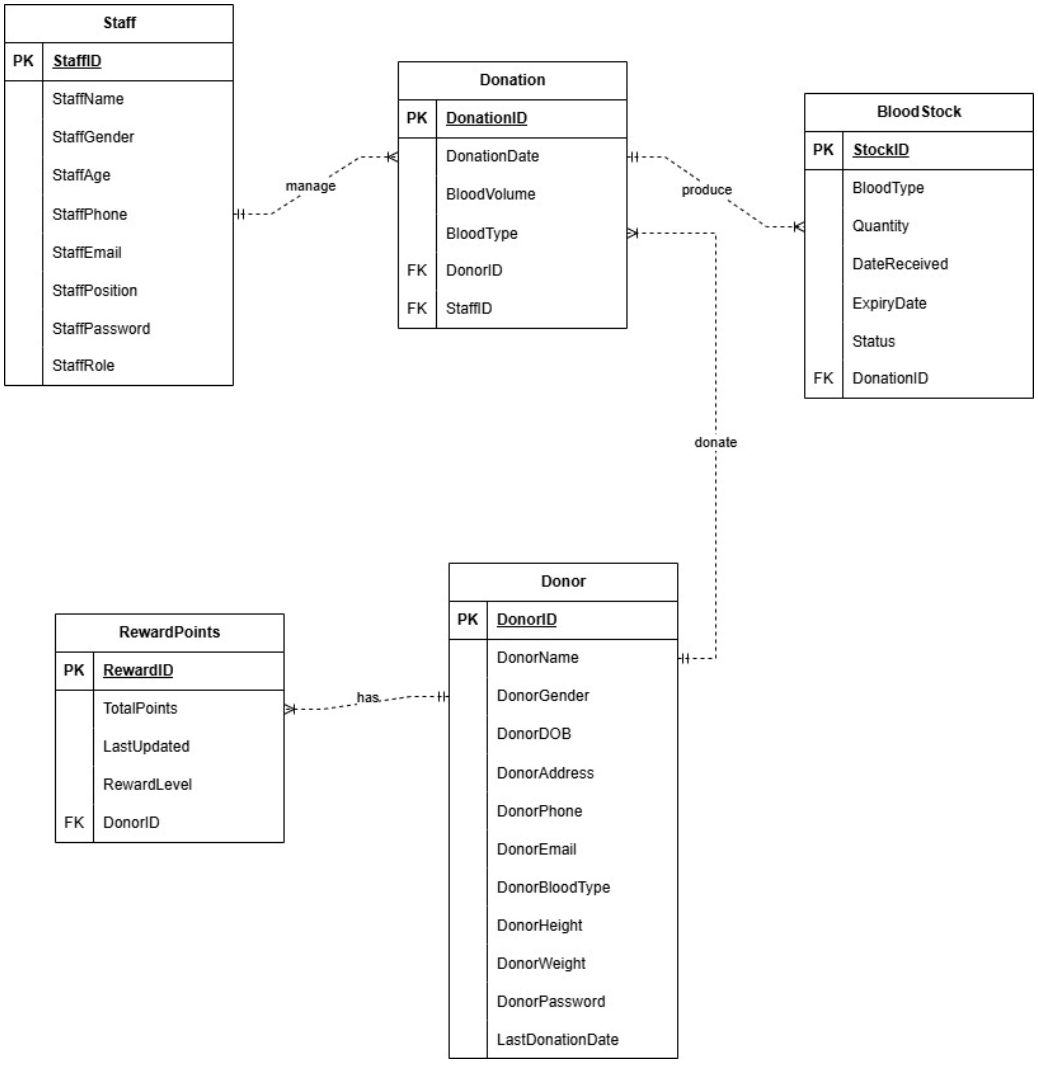



**Figure 3.6: Entity Relationship Diagram**

## 3.4 Data Normalization

**0NF (Unnormalized form)**
Raw data with duplicates and composite fields
Fields: (DonorID, DonorName, DonorGender, DonorDOB, DonorAddress, DonorPhone, DonorEmail, DonorBloodType, DonationID, DonationDate, BloodVolume, BloodType, StaffID, StaffName, StaffEmail, StaffPosition, RewardID, TotalPoints, RewardLevel, LastUpdated)

**1NF (First Normal Form)**

- Has a primary key
- No repeating group/ Multivalued attributes

Donor

| DonorID (PK) | DonorName | DonorGender | DonorBloodType |
|---|---|---|---|

Staff

| StaffID (PK) | StaffName |
|---|---|

Donation

| DonationID (PK) | DonorID | StaffID | DonationDate |
|---|---|---|---|

Reward Points

| RewardID (PK) | DonorID (FK) | RewardPoints |
|---|---|---|

**2NF (Second Normal Form)**

No partial dependency — all attributes depend on the full primary key

- Ensured that all non-key attributes in each table are fully functionally dependent on the whole primary key
- No composite primary keys with partial dependencies

Donor

| DonorID (PK) | DonorName | DonorGender | DonorBloodType |
|---|---|---|---|

Staff

| StaffID (PK) | StaffName |
|---|---|

Donation

| DonationID (PK) | DonorID (FK) | StaffID (FK) | DonationDate |
|---|---|---|---|

Reward Points

| RewardID (PK) | DonorID (FK) | RewardPoints |
|---|---|---|

**3NF (Third Normal Form)**

No transitive dependency — non-key attributes depend only on the primary key

- Removed any indirect dependencies between non-key attributes
- All attributes depend only on the primary key, not on other non-key fields

Donor

| DonorID (PK) | DonorName | DonorGender | DonorBloodType |
|---|---|---|---|

Staff

| StaffID (PK) | StaffName |
|---|---|

Donation

| DonationID (PK) | DonorID (FK to donor) | StaffID (FK to staff) | DonationDate |
|---|---|---|---|

Reward Points

| RewardID (PK) | DonorID (FK to donor) | RewardPoints |
|---|---|---|

## 3.5 Data Dictionary

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| StaffID | char(6) | NO | PRI | NULL | |
| StaffName | varchar(100) | NO | | NULL | |
| StaffGender | enum('Male','Female') | NO | | NULL | |
| StaffAge | int | NO | | NULL | |
| StaffPhone | varchar(15) | NO | | NULL | |
| StaffEmail | varchar(100) | NO | UNI | NULL | |
| StaffPosition | varchar(50) | NO | | NULL | |
| StaffPassword | varchar(255) | NO | | NULL | |
| StaffRole | enum('Admin','Staff') | NO | | Staff | |

**Figure 3.7: Data Dictionary for Table Staff**

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| DonorID | char(6) | NO | PRI | NULL | |
| DonorName | varchar(100) | NO | | NULL | |
| DonorGender | enum('Male','Female') | NO | | NULL | |
| DonorDOB | date | NO | | NULL | |
| DonorAge | int | NO | | NULL | |
| DonorAddress | varchar(255) | NO | | NULL | |
| DonorPhone | varchar(15) | NO | UNI | NULL | |
| DonorEmail | varchar(100) | NO | UNI | NULL | |
| DonorBloodType | enum('A','B','AB','O') | NO | | NULL | |
| DonorHeight | decimal(5,2) | NO | | NULL | |
| DonorWeight | decimal(5,2) | NO | | NULL | |
| DonorPassword | varchar(255) | NO | | NULL | |
| LastDonationDate | date | YES | | NULL | |

**Figure 3.8: Data Dictionary for Table Donor**

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| StockID | char(6) | NO | PRI | NULL | |
| BloodType | enum('A','B','AB','O') | NO | | NULL | |
| Quantity | int | NO | | 0 | |
| DateReceived | date | NO | | curdate() | DEFAULT_GENERATED |
| ExpiryDate | date | NO | | NULL | |
| Status | enum('Available','Used','Expired') | NO | | Available | |
| DonationID | char(6) | NO | MUL | NULL | |

**Figure 3.9: Data Dictionary for Table BloodStock**

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| DonationID | char(6) | NO | PRI | NULL | |
| DonationDate | date | NO | | curdate() | DEFAULT_GENERATED |
| BloodVolume | decimal(5,2) | NO | | NULL | |
| BloodType | enum('A','B','AB','O') | NO | | NULL | |
| DonorID | char(6) | NO | MUL | NULL | |
| StaffID | char(6) | NO | MUL | NULL | |

**Figure 3.10: Data Dictionary for Table Donation**

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| RewardID | char(6) | NO | PRI | NULL | |
| TotalPoints | int | NO | | 0 | |
| LastUpdated | datetime | NO | | CURRENT_TIMESTAMP | DEFAULT_GENERATED on update CURRENT_TI... |
| RewardLevel | enum('Bronze','Silver','Gold','Platinum') | NO | | Bronze | |
| DonorID | char(6) | NO | UNI | NULL | |

**Figure 3.11: Data Dictionary for Table RewardPoints**

## 3.6 Interface Design

### 3.6.1 Main Page

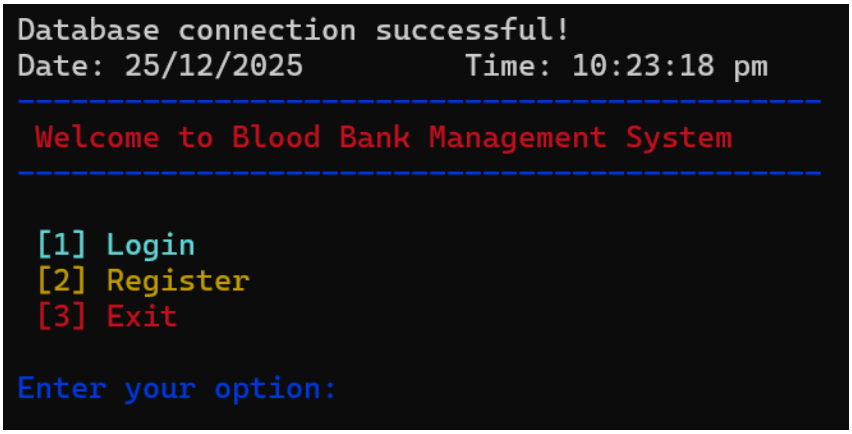



**Figure 3.12: Main Page Interface**

### 3.6.2 Login Menu Page

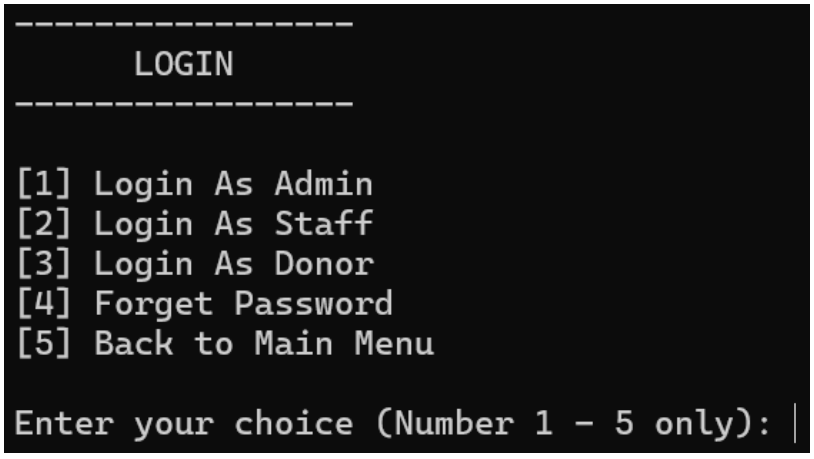



**Figure 3.13: Login Menu Interface**

### 3.6.3 Admin Access Page

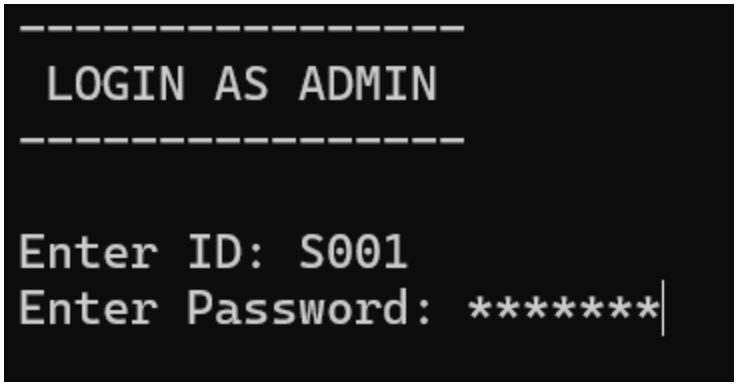



**Figure 3.14: Admin Access Interface**

### 3.6.4 Admin Main Menu

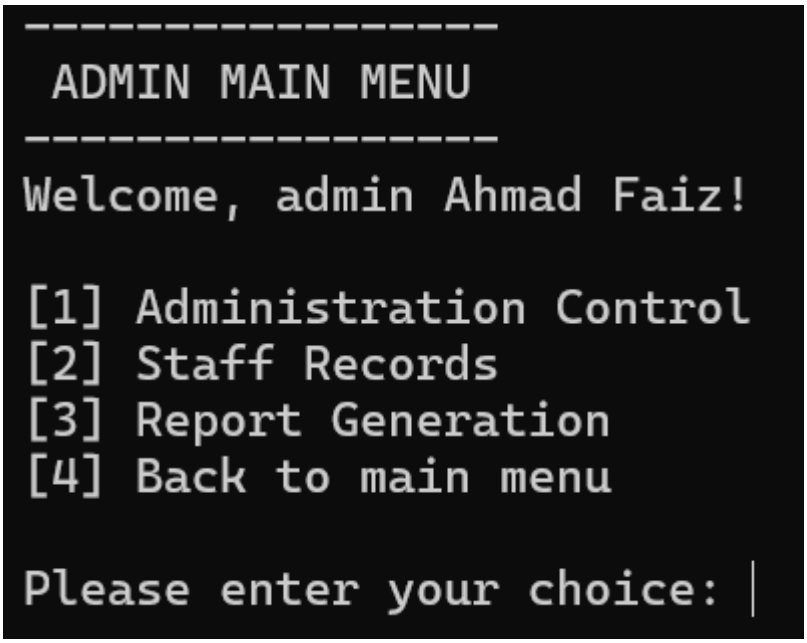



**Figure 3.15: Admin Main Menu Interface**

### 3.6.5 Admin Control Menu

```
--------------------
 ADMIN CONTROL MENU
--------------------
Welcome, admin! What would you like to do with Staff?

[1] Add Staff Record
[2] Delete Staff Record
[3] Update Staff Record
[4] Search Staff Record
[5] Main Menu

Please enter your choice (1, 2, 3, 4, 5):
```

**Figure 3.16: Admin Control Interface**

### 3.6.6 Admin Add Staff

```
--------------------
 ADMIN ADD - STAFF
--------------------
New Staff ID: S013
Name: Jordan
Gender: Male
Age: 37
Phone Number: 0123456777
Email: jordan@gmail.com
Position: Nurse
Staff Password: pass013
Staff Role: Staff

A new staff record was successfully added!

Do you want to add another record? (Y/N):
```

**Figure 3.17: Admin Add Staff**

### 3.6.7 Admin Delete Staff

```
----------------------
   ADMIN DELETE MENU
----------------------

*** EXISTING STAFF LIST ***
-------------------------------------------------------------------------------------------------------------------
ID        Name               Gender  Age   Phone          Email                    Position              Role
-------------------------------------------------------------------------------------------------------------------
S001      Ahmad Faiz         Male    34    0123456789     ahmadfaiz@gmail.com      Doctor                Admin
S002      Nurul Aisyah       Female  29    0198765432     nurulaisyah@yahoo.com    Nurse                 Admin
S003      Lim Wei Ming       Male    31    0171234567     limweiming@gmail.com     Lab Technician        Staff
S004      Tan Mei Ling       Female  28    0162345678     tanmeiling@yahoo.com     Nurse                 Staff
S005      Rajesh Kumar       Male    36    0123456790     rajeshkumar@gmail.com    Doctor                Admin
S006      Siti Hajar         Female  32    0191234567     sitihajar@yahoo.com      Lab Technician        Staff
S007      Chong Li Hua       Female  30    0172345678     chonglihua@gmail.com     Nurse                 Staff
S008      Arun Prakash       Male    35    0163456789     arunprakash@yahoo.com    Doctor                Admin
S009      Lee Chee Keong     Male    33    01245678890    leechkeong@gmail.com     Lab Technician        Staff
S010      Aishah Rahman      Female  27    0192345678     aishahrahman@yahoo.com   Nurse                 Staff
S011      Chin Weng Leong    Male    57    01233337777    chin@gmail.com           Doctor                Staff
S012      Susan              Female  43    0123456788     susan1@gmail.com         Nurse                 Staff
S013      Kathy              Female  56    0123456798     kathy@gmail.com          Doctor                Staff

Enter Staff ID to delete: S013

*** STAFF TO DELETE ***
Staff ID: S013
Name: Kathy
Gender: Female
Age: 56
Phone: 0123456798
Email: kathy@gmail.com
Position: Doctor
Role: Staff

Are you sure you want to delete this record? [Y/N]: Y

Successfully deleted the staff.
Press any key to continue . . .
```

**Figure 3.18: Admin Delete Staff**

### 3.6.8 Admin Update Staff

```
----------------------
 ADMIN UPDATE - STAFF
----------------------

================================================== STAFF LIST ==================================================
ID       NAME                  GENDER    AGE       PHONE          EMAIL                    POSITION          ROLE
------------------------------------------------------------------------------------------------------------------

S001     Ahmad Faiz            Male      34        0123456789     ahmadfaiz@gmail.com      Doctor            Admin
S002     Nurul Aisyah          Female    29        0198765432     nurulaisyah@yahoo.com    Nurse             Admin
S003     Lim Wei Ming          Male      31        0171234567     limweiming@gmail.com     Lab Technician    Staff
S004     Tan Mei Ling          Female    28        0162345678     tanmeiling@yahoo.com     Nurse             Staff
S005     Rajesh Kumar          Male      36        0123456790     rajeshkumar@gmail.com    Doctor            Admin
S006     Siti Hajar            Female    32        0191234567     sitihajar@yahoo.com      Lab Technician    Staff
S007     Chong Li Hua          Female    30        0172345678     chonglihua@gmail.com     Nurse             Staff
S008     Arun Prakash          Male      35        0163456789     arunprakash@yahoo.com    Doctor            Admin
S009     Lee Chee Keong        Male      33        0124567890     leechkeong@gmail.com     Lab Technician    Staff
S010     Aishah Rahman         Female    27        0192345678     aishahrahman@yahoo.com   Nurse             Staff
S011     Chin Weng Leong       Male      57        01233337777    chin@gmail.com           Doctor            Staff
S012     Susan                 Female    43        0123456788     susan1@gmail.com         Nurse             Staff
S013     Jordan                Male      37        0123456777     jordan@gmail.com         Nurse             Staff
==================================================================================================================

Enter staff ID to update: S013

Current Info:
Name: Jordan
Gender: Male
Age: 37
Phone: 0123456777
Email: jordan@gmail.com
Position: Nurse
Role: Staff

Which field do you want to update?
[1] Name
[2] Gender
[3] Age
[4] Phone
[5] Email
[6] Position
[7] Password
[8] Role

Your choice: 3
New Age: 38

Successfully Updated!
Do you want to continue updating another field? [Y/N]:
```

**Figure 3.19: Admin Update Staff**

### 3.6.9 Admin Search Staff

```
----------------------
 ADMIN SEARCH - STAFF
----------------------

Search by:
[1] Staff ID
[2] Name
[3] Gender
[4] Age
[5] Phone
[6] Email
[7] Position
[8] Role
[9] Back to main menu

Your choice >>
```

**Figure 3.20: Admin Search Staff**

### 3.6.10 Admin Display Staff Record

```
--------------
 STAFF RECORD
--------------
Here is the record of staff:

-------------------------------------------------------------------------------------------------------------------------
Staff ID    Name              Gender    Age   Phone          Email                   Position          Role
-------------------------------------------------------------------------------------------------------------------------
S001        Ahmad Faiz        Male      34    0123456789     ahmadfaiz@gmail.com      Doctor            Admin
S002        Nurul Aisyah      Female    29    0198765432     nurulaisyah@yahoo.com    Nurse             Admin
S003        Lim Wei Ming      Male      31    0171234567     limweiming@gmail.com     Lab Technician    Staff
S004        Tan Mei Ling      Female    28    0162345678     tanmeiling@yahoo.com     Nurse             Staff
S005        Rajesh Kumar      Male      36    0123456790     rajeshkumar@gmail.com    Doctor            Admin
S006        Siti Hajar        Female    32    0191234567     sitihajar@yahoo.com      Lab Technician    Staff
S007        Chong Li Hua      Female    30    0172345678     chonglihua@gmail.com     Nurse             Staff
S008        Arun Prakash      Male      35    0163456789     arunprakash@yahoo.com    Doctor            Admin
S009        Lee Chee Keong    Male      33    0124567890     leechkeong@gmail.com     Lab Technician    Staff
S010        Aishah Rahman     Female    27    0192345678     aishahrahman@yahoo.com   Nurse             Staff
S011        Chin Weng Leong   Male      57    01233337777    chin@gmail.com           Doctor            Staff
S012        Susan             Female    43    0123456788     susan1@gmail.com         Nurse             Staff
-------------------------------------------------------------------------------------------------------------------------
Press any key to continue . . .
```

**Figure 3.21: Admin Display Staff Record Interface**

### 3.6.11 Admin Generate Report

```
Date: 25/12/2025  Time:        11:07:16 pm
-----------------------------
 REPORT GENERATION FOR ADMIN
-----------------------------
Total number of staff in the system: 12

Which attribute do you want to generate as report?
[1] Gender
[2] Position
[3] Age
[4] Back to Main Menu
Your choice:
```

**Figure 3.22: Admin Generate Report Interface**

### 3.6.12 Admin Generate Report By Gender

```
Date: 28/12/2025  Time:        03:02:55 pm

Staff count by gender:
Male = 6
Female = 6

**************************************************
|         Graph (GENDER VS NUMBER OF STAFF)        |
**************************************************

Male    |****** 6
Female  |****** 6

Do you want to export this report to a TXT file? (Y/N):
```

**Figure 3.23: Admin Generate Report By Gender**

### 3.6.13 Admin Generate Report By Position

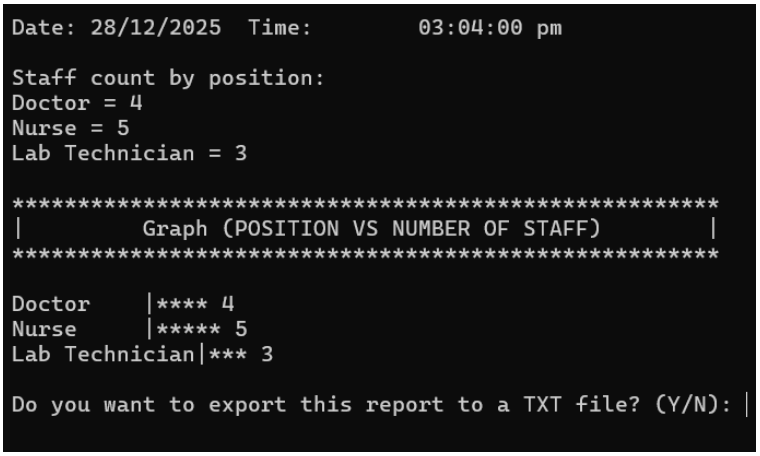



**Figure 3.24: Admin Generate Report By Position**

### 3.6.14 Admin Generate Report By Age

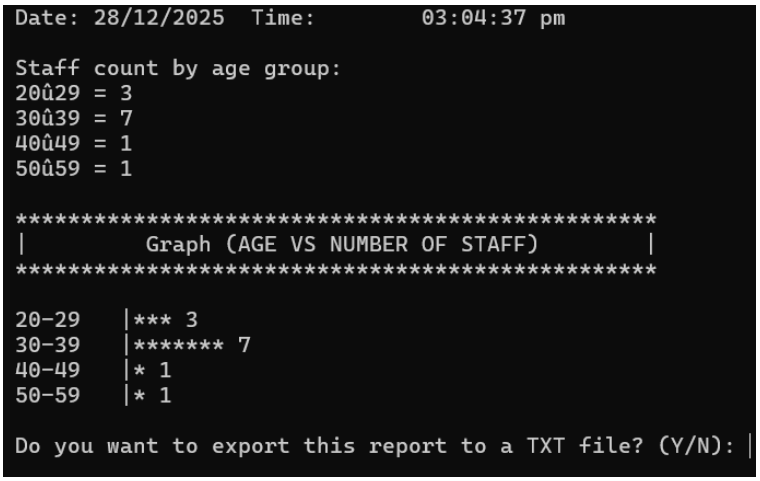



**Figure 3.25: Admin Generate Report By Age**

### 3.6.15 Staff Access Page

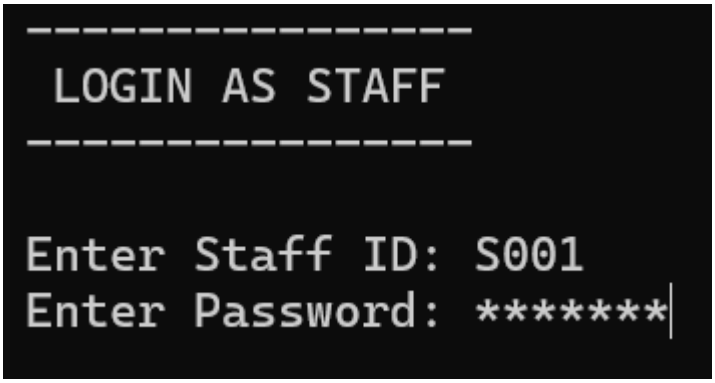



**Figure 3.26: Staff Access Interface**

### 3.6.16 Staff Main Menu

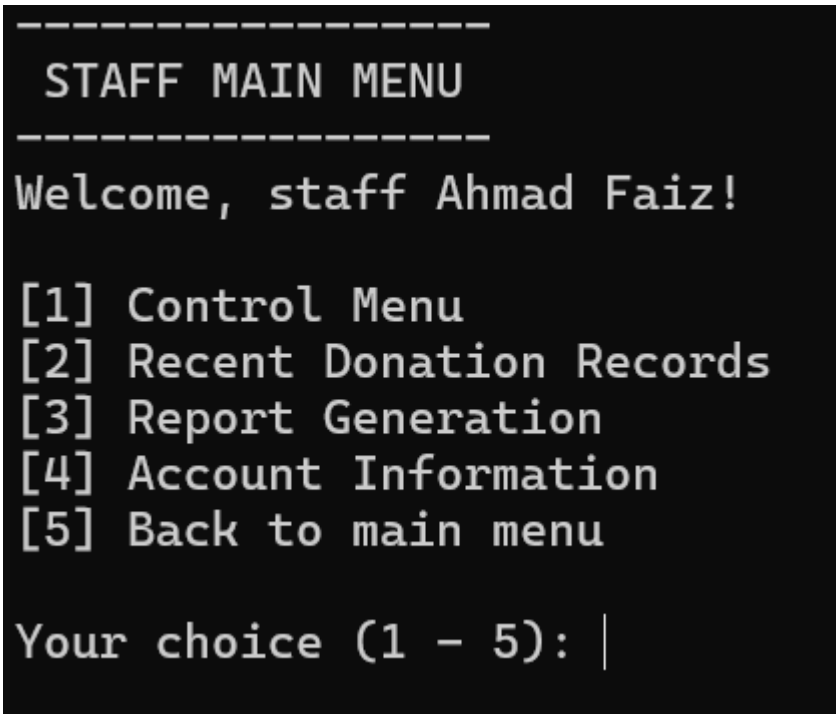



**Figure 3.27: Staff Main Menu Interface**

### 3.6.17 Staff Control Menu

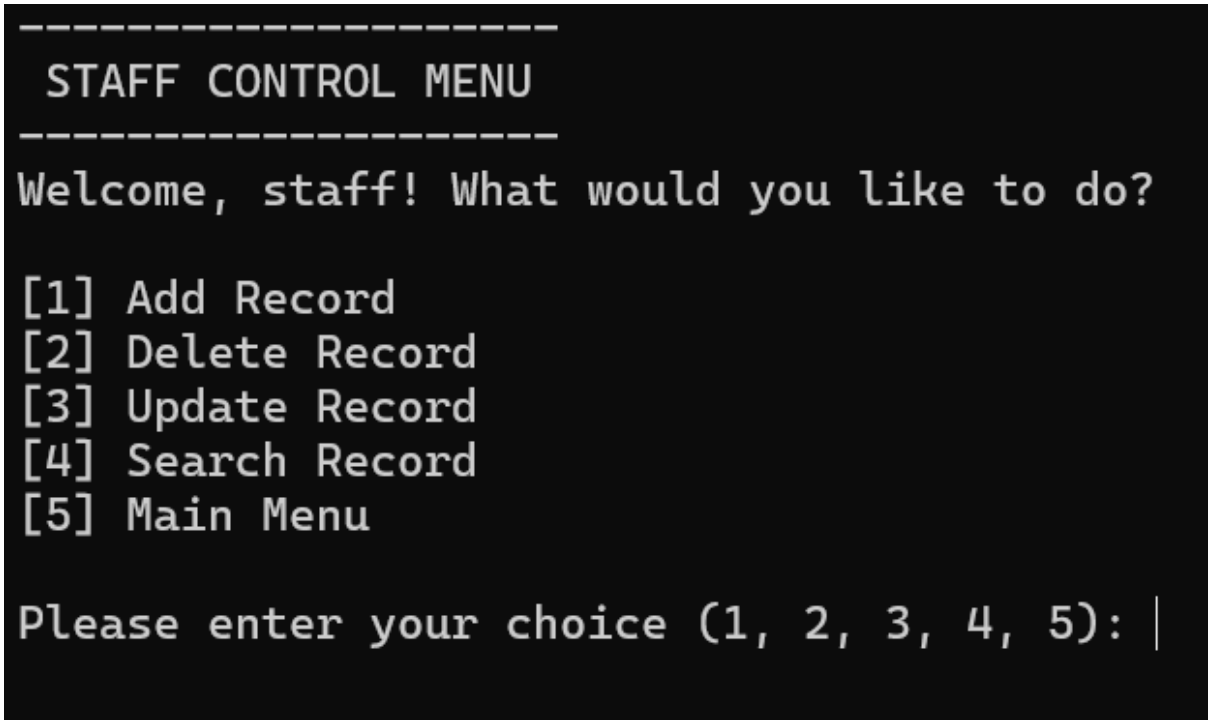



**Figure 3.28: Staff Control Menu Interface**

### 3.6.18 Staff Add Record Menu

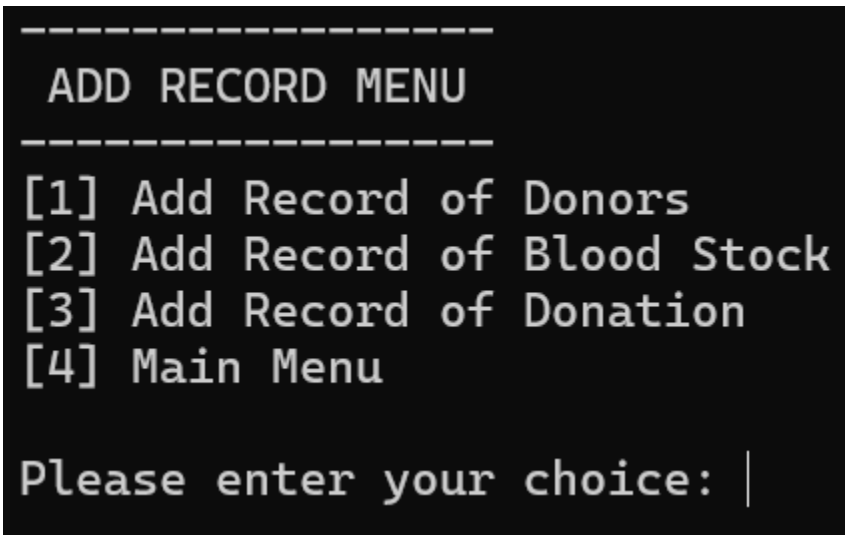



**Figure 3.29: Staff Add Record Menu Interface**

### 3.6.19 Staff Add Record Menu-Add Donor

```
--------------------
 ADD RECORDS - DONOR
--------------------
New Donor ID: D013

Name: Jordan
Gender (Male/Female): Male
Date of Birth (YYYY-MM-DD): 1987-4-12
Address: 12 Jalan Meriah Melaka
Blood Type (A/B/AB/O): B
Height (cm): 178
Weight (kg): 78
Phone Number: 0123456787
Email: jordan@gmail.my
Password: *********

Donor successfully added to the database!
Do you want to continue adding records? [Y/N]:
```

**Figure 3.30: Staff Add Record Menu-Add Donor**

### 3.6.20 Staff Add Record Menu-Add Block Stock

```
---------------------------
 ADD RECORDS - BLOOD STOCK
---------------------------
New Stock ID: BS013
Blood Type (A/B/AB/O): B
Quantity: 470
Expiry Date (YYYY-MM-DD): 2025-12-25
Donation ID (e.g. DN***): DN013

Blood stock successfully added to the database.
Do you want to continue adding records? [Y/N]:
```

**Figure 3.31: Staff Add Record Menu-Add Block Stock**

### 3.6.21 Staff Add Record Menu-Add Donation

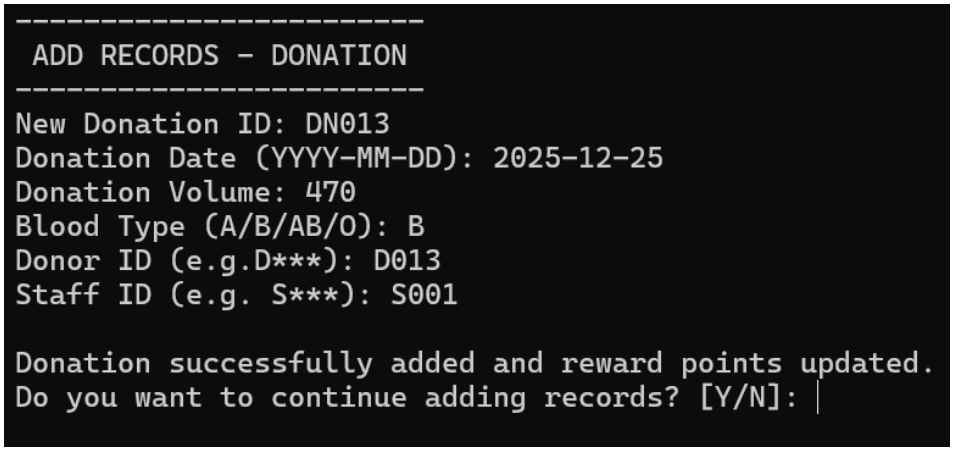



**Figure 3.32: Staff Add Record Menu-Add Donation**

### 3.6.22 Staff Delete Record Menu

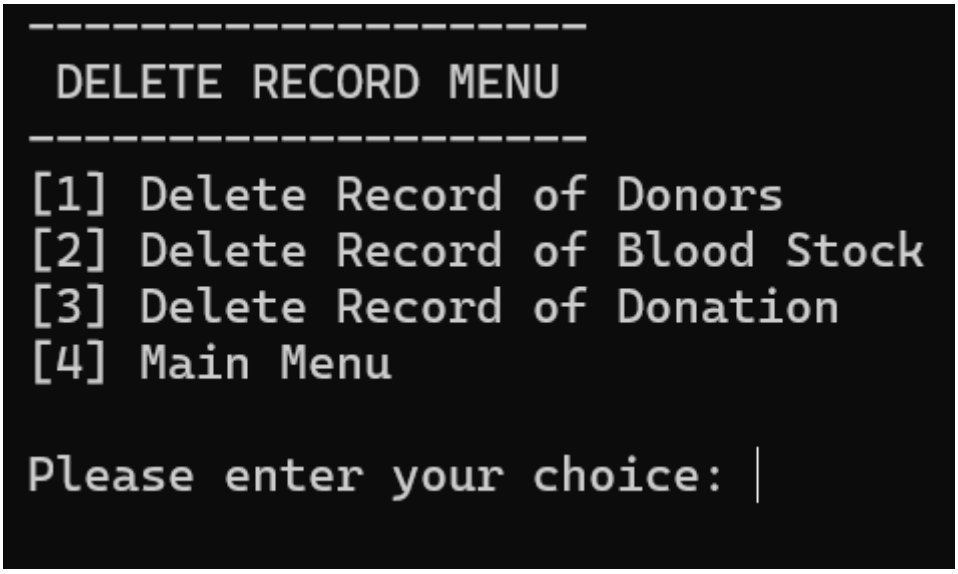



**Figure 3.33: Staff Delete Record Menu Interface**

### 3.6.23 Staff Delete Record Menu-Delete Donor Record

```
----------------------
 DELETE RECORD - DONOR
----------------------

========================== ALL DONORS ==========================
ID        NAME                    BLOOD TYPE     PHONE
----------------------------------------------------------------
D001      Mohd Hafiz              O              0123451111
D002      Siti Sarah              A              0198762222
D003      Goh Wei Jie             B              0171233333
D004      Lim Mei Fang            AB             0162344444
D005      Ravi Kumar              O              0123455555
D006      Aina Farhana            A              0191236666
D007      Chong Li Mei            B              0172347777
D008      Arjun Singh             AB             0163458888
D009      Lee Jun Hao             O              0124569999
D010      Nur Hidayah             A              0192340000
D011      Albie Yoke              O              0164456888
D012      John                    O              0123456777
D013      Jordan                  B              0123456787
================================================================

Enter Donor ID to delete: D013

Here's the record found:

Donor ID: D013
Name: Jordan
Gender: Male
Age: 38
Date of Birth: 1987-04-12
Address: 12 Jalan Meriah Melaka
Blood Type: B
Height: 178
Weight: 78
Phone: 0123456787
Email: jordan@gmail.com
Last Donation Date: 2025-12-25

Are you sure you want to delete this record [Y/N]: Y

Successfully deleted the donor!
Do you want to continue deleting? [Y/N]:
```

**Figure 3.34: Staff Delete Record Menu-Delete Donor Record**

### 3.6.24 Staff Delete Record Menu-Delete Blood Stock Record

```
----------------------------
 DELETE RECORD - BLOOD STOCK
----------------------------

================ ALL BLOOD STOCK ================

StockID     Type        Qty       Received       Expiry         Status         DonationID
-------------------------------------------------------------------------------
BS001       A           450       2025-01-15     2025-02-26     Available      DN001
BS002       A           500       2025-03-20     2025-05-01     Available      DN002
BS003       B           470       2025-02-10     2025-03-24     Available      DN003
BS004       AB          450       2025-04-05     2025-05-17     Available      DN004
BS005       O           500       2025-01-25     2025-03-07     Available      DN005
BS006       A           450       2025-03-10     2025-04-21     Available      DN006
BS007       B           480       2025-02-28     2025-04-11     Available      DN007
BS008       AB          500       2025-03-15     2025-04-26     Available      DN008
BS009       O           470       2025-01-30     2025-03-11     Available      DN009
BS010       A           450       2025-04-12     2025-05-24     Available      DN010
BS011       O           500       2025-11-30     2026-04-27     Available      DN011
BS012       O           480       2025-11-30     2026-04-24     Available      DN012
BS013       B           470       2025-12-25     2025-12-25     Available      DN013

Enter Blood Stock ID to delete: BS013

Here's the record found:

Stock ID     : BS013
Blood Type   : B
Quantity     : 470
Date Received: 2025-12-25
Expiry Date  : 2025-12-25
Status       : Available
Donation ID  : DN013

Are you sure to delete this record [Y/N]: Y

Successfully deleted the blood stock!

Do you want to continue deleting record? [Y/N]:
```

**Figure 3.35: Staff Delete Record Menu-Delete Blood Stock Record**

### 3.6.25 Staff Delete Record Menu-Delete Donation Record

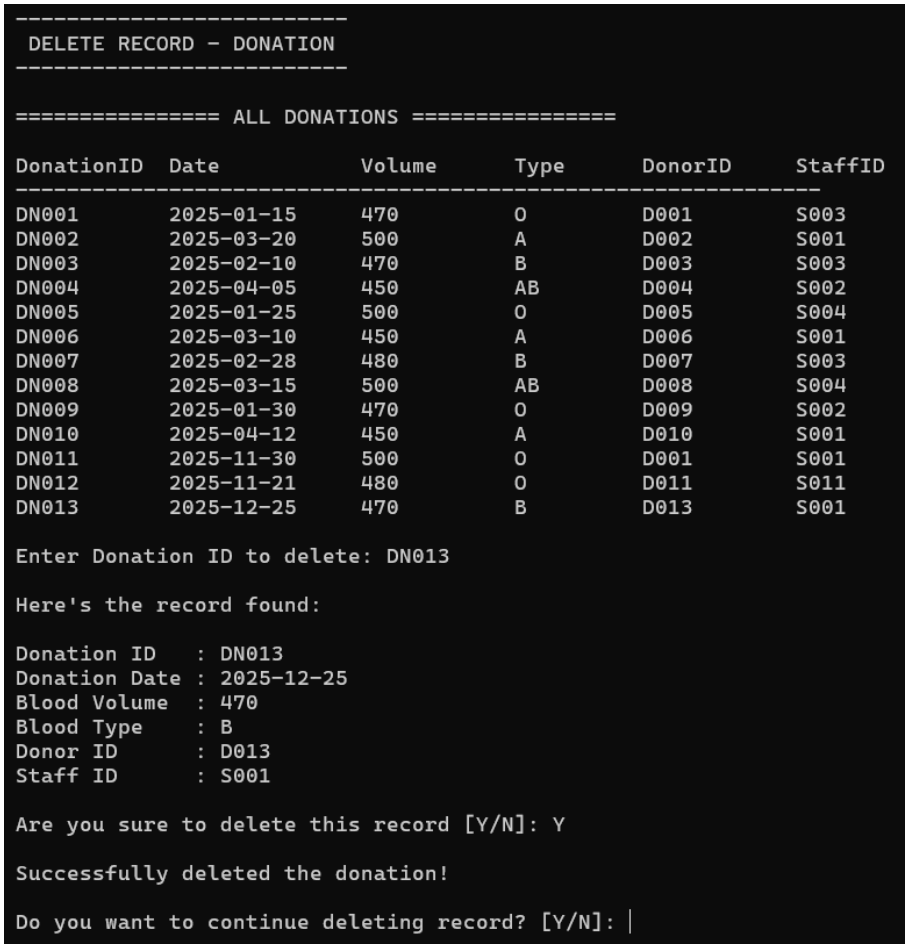



**Figure 3.36: Staff Delete Record Menu-Delete Donation Record**

### 3.6.26 Staff Update Record Menu

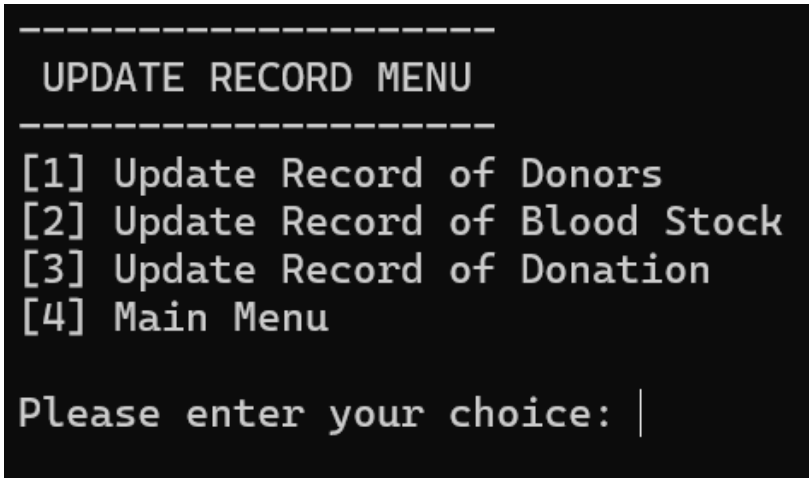



**Figure 3.37: Staff Update Record Menu Interface**

### 3.6.27 Staff Update Record Menu-Update Donor

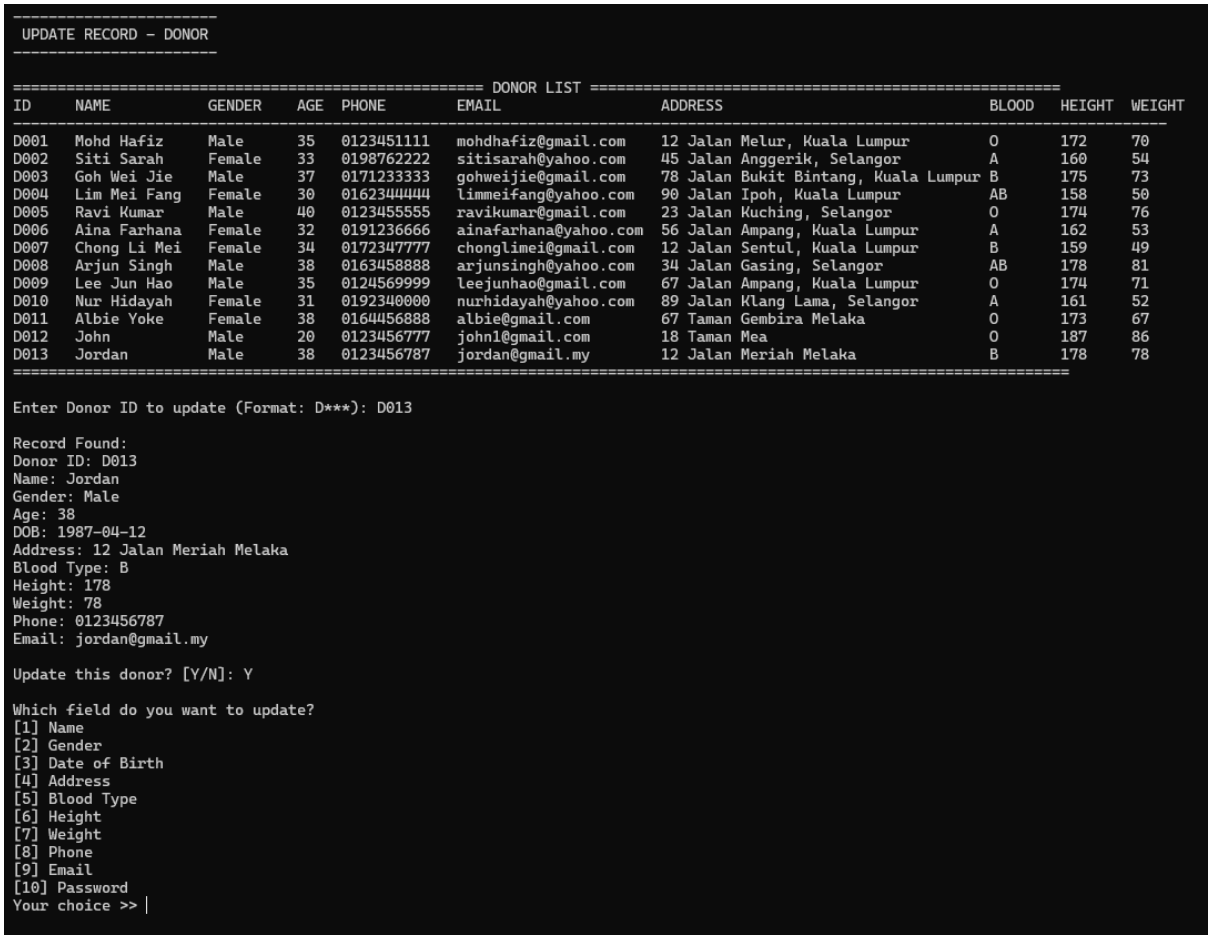



**Figure 3.38: Staff Update Record Menu-Update Donor**

### 3.6.28 Staff Update Record Menu-Update Blood Stock

```
-----------------------------
 UPDATE RECORD - BLOOD STOCK
-----------------------------

================================ BLOOD STOCK LIST ================================
StockID     Type        Qty       Received       Expiry         Status         DonationID
--------------------------------------------------------------------------------------
BS001       A           450       2025-01-15     2025-02-26     Available      DN001
BS002       A           500       2025-03-20     2025-05-01     Available      DN002
BS003       B           470       2025-02-10     2025-03-24     Available      DN003
BS004       AB          450       2025-04-05     2025-05-17     Available      DN004
BS005       O           500       2025-01-25     2025-03-07     Available      DN005
BS006       A           450       2025-03-10     2025-04-21     Available      DN006
BS007       B           480       2025-02-28     2025-04-11     Available      DN007
BS008       AB          500       2025-03-15     2025-04-26     Available      DN008
BS009       O           470       2025-01-30     2025-03-11     Available      DN009
BS010       A           450       2025-04-12     2025-05-24     Available      DN010
BS011       O           500       2025-11-30     2026-04-27     Available      DN011
BS012       O           480       2025-11-30     2026-04-24     Available      DN012
BS013       B           470       2025-12-25     2025-12-25     Available      DN013
======================================================================================

Enter Blood Stock ID to search (Format: BS***): BS013

Here's the record found:

Stock ID: BS013
Blood Type: B
Quantity: 470
Date Received: 2025-12-25
Expiry Date: 2025-12-25
Status: Available
Donation ID: DN013

Are you sure to update this record? [Y/N]: Y

Select information to update:
[1] Blood Type
[2] Quantity
[3] Expiry Date
[4] Status

Your choice >>
```

**Figure 3.39: Staff Update Record Menu-Update Blood Stock**

### 3.6.29 Staff Update Record Menu-Update Donation

```
-------------------------
 UPDATE RECORD - DONATION
-------------------------

============================== DONATION LIST ==============================
DonationID  Date          Volume      Type        DonorID     StaffID
---------------------------------------------------------------------------
DN001       2025-01-15    470         O           D001        S003
DN002       2025-03-20    500         A           D002        S001
DN003       2025-02-10    470         B           D003        S003
DN004       2025-04-05    450         AB          D004        S002
DN005       2025-01-25    500         O           D005        S004
DN006       2025-03-10    450         A           D006        S001
DN007       2025-02-28    480         B           D007        S003
DN008       2025-03-15    500         AB          D008        S004
DN009       2025-01-30    470         O           D009        S002
DN010       2025-04-12    450         A           D010        S001
DN011       2025-11-30    500         O           D001        S001
DN012       2025-11-21    480         O           D011        S011
DN013       2025-12-25    470         B           D013        S001
===========================================================================

Enter Donation ID to search (Format: DN***): DN013
Donation ID: DN013
Donation Date: 2025-12-25
Blood Volume: 470
Blood Type: B
Donor ID: D013
Staff ID: S001

Do you want to update this record? [Y/N]: Y
----------------------------
  Updation of Information
----------------------------
[1] Blood Volume
[2] Blood Type
[3] Donor ID
[4] Staff ID
Your choice >>
```

**Figure 3.40: Staff Update Record Menu-Update Donation**

### 3.6.30 Staff Search Record Menu

```
--------------------
 SEARCH RECORD MENU
--------------------
[1] Search Record of Donors
[2] Search Record of Blood Stock
[3] Search Record of Donation
[4] Main Menu

Please enter your choice:
```

**Figure 3.41: Staff Search Record Menu Interface**

### 3.6.31 Staff Search Record Menu-Staff Search Donor Record

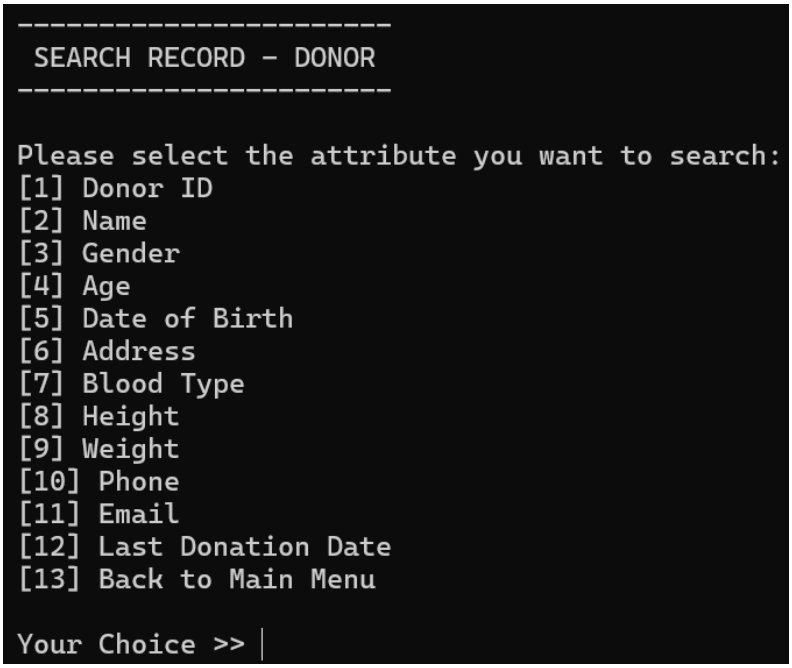



**Figure 3.42: Staff Search Record Menu-Staff Search Donor Record**

### 3.6.32 Staff Search Record Menu-Staff Search Blood Stock Record

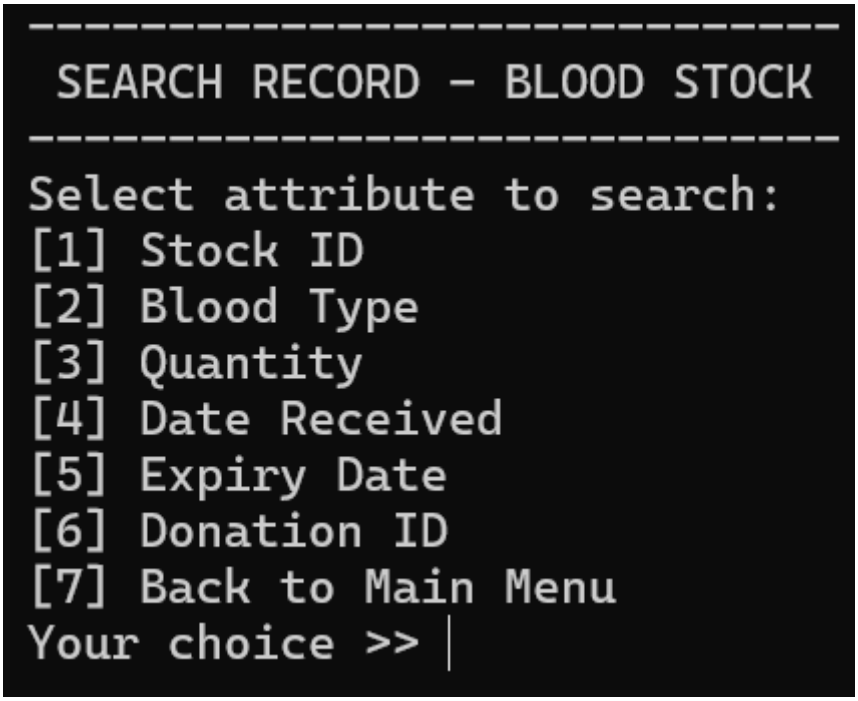



**Figure 3.43: Staff Search Record Menu-Staff Search Blood Stock Record**

### 3.6.33 Staff Search Record Menu-Staff Search Donation Record

```
--------------------------
 SEARCH RECORD - DONATION
--------------------------
Please select the attribute you want to search:
[1] Donation ID
[2] Donation Date
[3] Donor ID
[4] Staff ID
[5] Blood Type
[6] Blood Volume
[7] Back to Main Menu

Your Choice >>
```

**Figure 3.44: Staff Search Record Menu-Staff Search Donation Record**

### 3.6.34 Blood Donation Report Page

```
==========================================
            BLOOD DONATION REPORT
==========================================
Generated on: Thursday, 12/25/2025 11:59:36 PM

Currently, there are 13 donors registered in the system.

---------- REPORT DATE RANGE ----------
Start Year (YYYY): 2024
Start Month (MM): 2
Start Day (DD): 12

End Year (YYYY): 2025
End Month (MM): 2
End Day (DD): 12

Results:
-----------------------------------------------------------------------------------------
| Record     | Donor Name               | Date         | Volume       | Type     |
-----------------------------------------------------------------------------------------
| DN001      | Mohd Hafiz               | 2025-01-15   | 470.00       | O        |
| DN003      | Goh Wei Jie              | 2025-02-10   | 470.00       | B        |
| DN005      | Ravi Kumar               | 2025-01-25   | 500.00       | O        |
| DN009      | Lee Jun Hao              | 2025-01-30   | 470.00       | O        |
-----------------------------------------------------------------------------------------

Total donations found: 4 between 2024-2-12 and 2025-2-12.

----------------------------------------
Would you like to export this report to a TXT file? (Y/N):
```

**Figure 3.45: Blood Donation Report Interface**

### 3.6.35 Staff Account Information Page

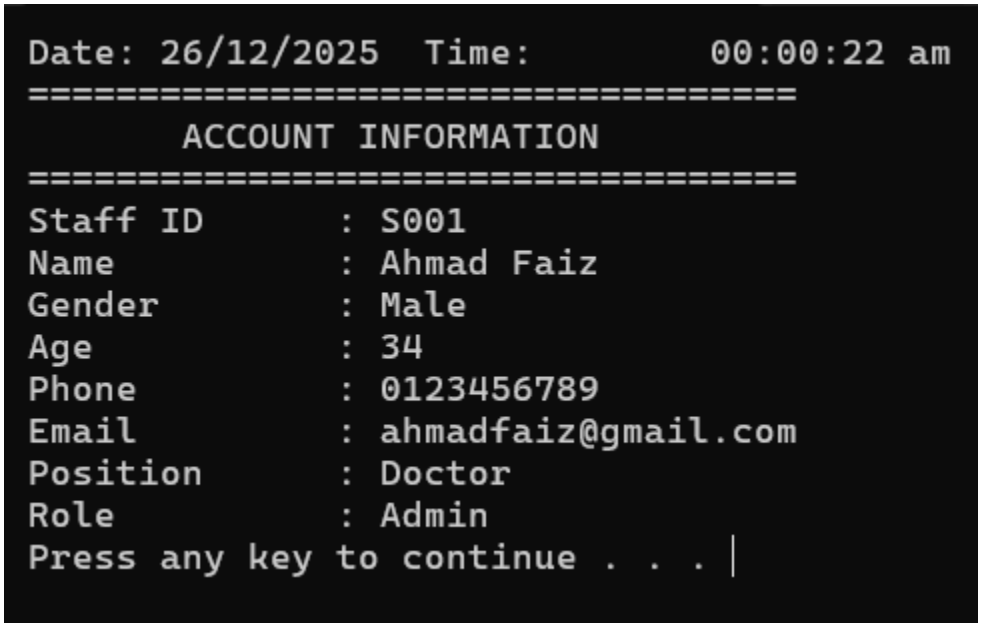



**Figure 3.46: Staff Account Information Interface**

### 3.6.36 Donor Access Page

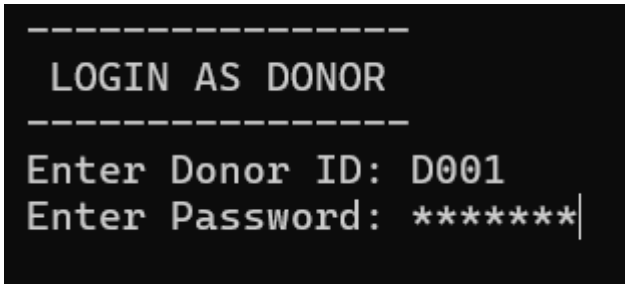



**Figure 3.47: Donor Access Interface**

### 3.6.37 Donor Main Menu

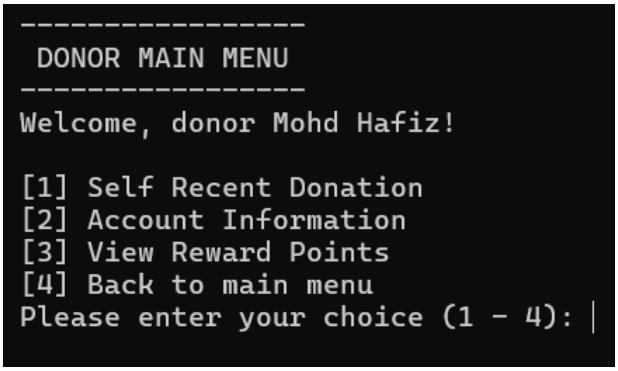



**Figure 3.48: Donor Main Menu**

### 3.6.38 Donor Recent Donation Record Page

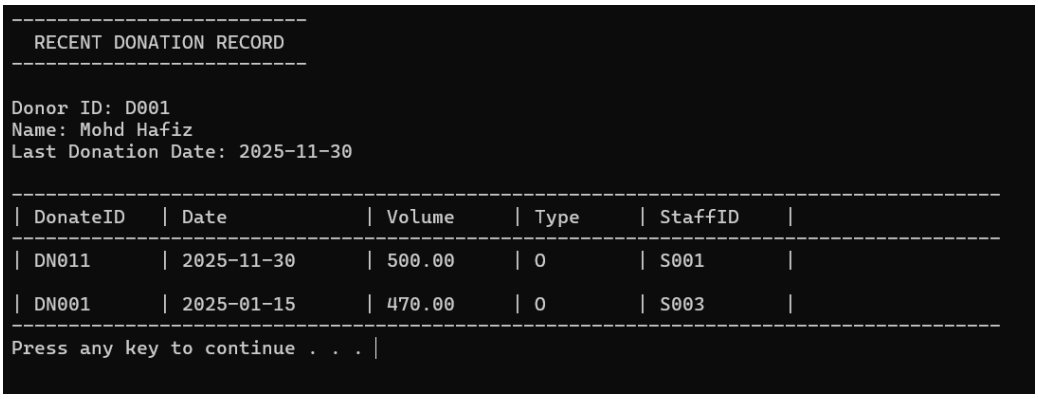



**Figure 3.49: Donor Recent Donation Record Interface**

### 3.6.39 Donor Account Information Page




**Figure 3.50: Donor Account Information Interface**

### 3.6.40 Donor Reward Points

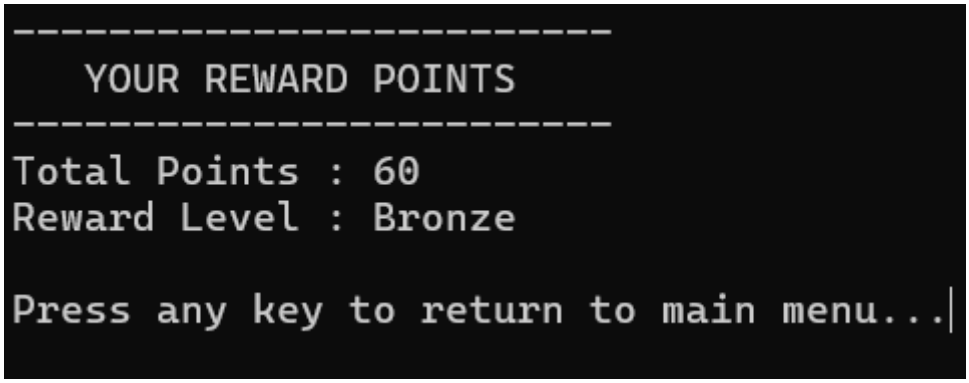



**Figure 3.51: Donor Reward Points Interface**

### 3.6.41 Account Registration Menu Page

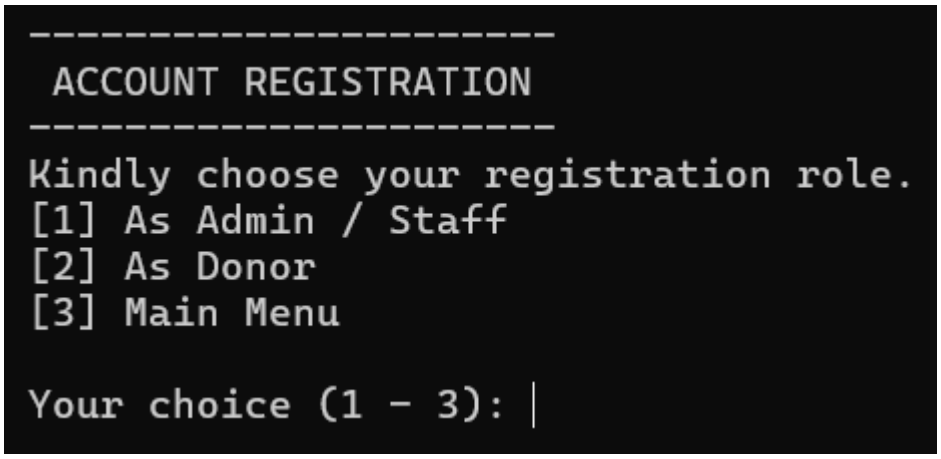



**Figure 3.52: Account Registration Menu Interface**

### 3.6.42 New Admin/Staff Registration

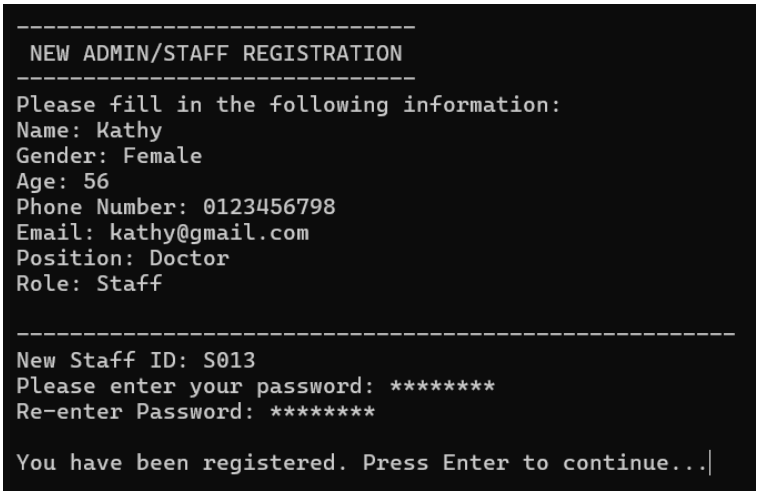



**Figure 3.53: New Admin/Staff Registration**

### 3.6.43 New Donor Registration

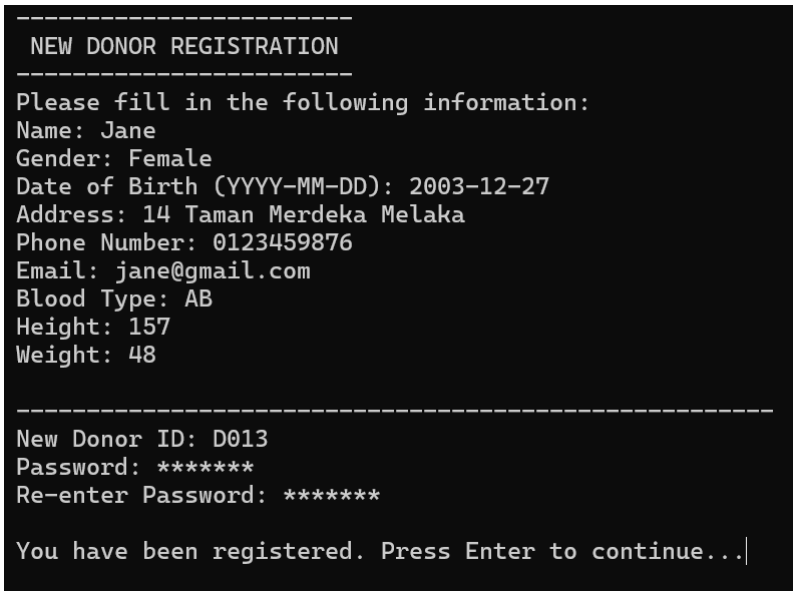



**Figure 3.54: New Donor Registration**

### 3.6.44 Password Recovery Menu Page

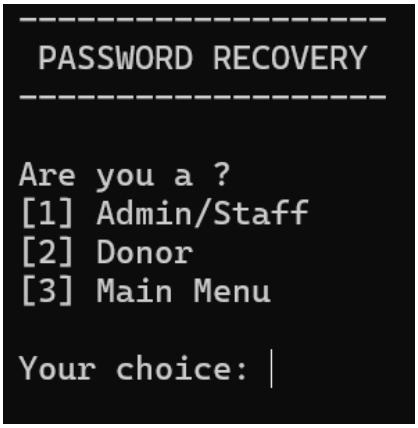



**Figure 3.55: Password Recovery Menu Interface**

### 3.6.45 Admin/Staff Password Recovery

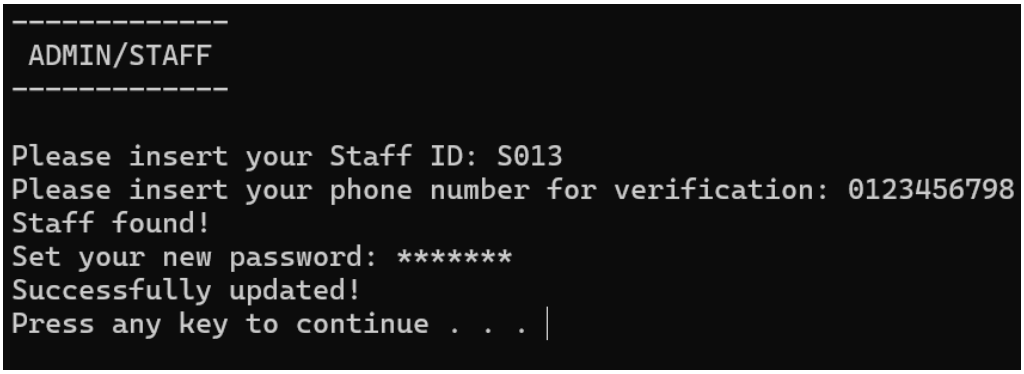



**Figure 3.56: Admin/Staff Password Recovery**

### 3.6.46 Donor Recovery

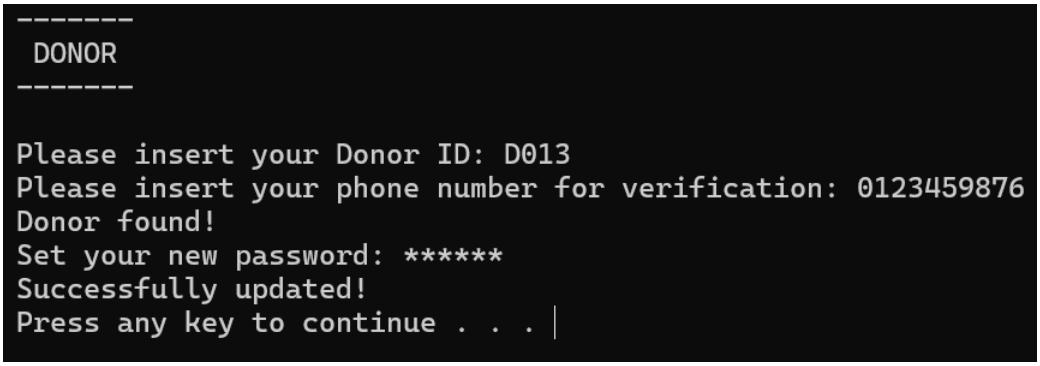



**Figure 3.57: Donor Recovery**

# 4 IMPLEMENTATION

## 4.6 Naming Convention

Naming Convention in programming are standard rules or guidelines for naming things like variables, functions, classes, files, and constants. They make code easier to read, understand, and maintain. Here are some key elements of naming conventions in programming:

### 4.6.1 Pascal Case

For Pascal Case, every word starts with an uppercase letter. For example, StaffID, StaffName, StaffGender, StaffAge, StaffPhone, StaffEmail, StaffPosition, StaffRole.

```
cout << "\n*** STAFF TO DELETE ***\n";
cout << "Staff ID: " << res->getString("StaffID") << endl;
cout << "Name: " << res->getString("StaffName") << endl;
cout << "Gender: " << res->getString("StaffGender") << endl;
cout << "Age: " << res->getInt("StaffAge") << endl;
cout << "Phone: " << res->getString("StaffPhone") << endl;
cout << "Email: " << res->getString("StaffEmail") << endl;
cout << "Position: " << res->getString("StaffPosition") << endl;
cout << "Role: " << res->getString("StaffRole") << endl;
```

**Figure 4.1: Pascal Case Implementation**

## 4.7 Function

A function is a reusable block of code that performs a specific task, which helps organize a program into smaller, manageable parts. It can accept input values called parameters, process them, and optionally return a result, making programs easier to read, test, and maintain while avoiding repeated code. For example, MainMenu function used to display the title as well as login, register and exit options.

```
void MainMenu()
{
    system("cls");
    testDBConnection();
    char optionMain;
    ShowTime();
    cout << foreground(Blue) << "-----------------------------------------------" << resetANSI() << endl;
    cout << foreground(Red) << " Welcome to Blood Bank Management System " << resetANSI() << endl;
    cout << foreground(Blue) << "-----------------------------------------------" << resetANSI() << endl;
    cout << endl;

    cout << foreground(Light_Aqua) << " [1] Login " << resetANSI() << endl;
    cout << foreground(Yellow) << " [2] Register " << resetANSI() << endl;
    cout << foreground(Red) << " [3] Exit " << resetANSI() << endl;
    cout << endl;

    cout << foreground(Blue) << "Enter your option: " << resetANSI();
    cin >> optionMain;

    while (optionMain != '1' && optionMain != '2' && optionMain != '3' && optionMain != '4')
    {
        cout << "Invalid Choice. Please enter again: " << endl;
        cin >> optionMain;
    }

    if (optionMain == '1') MainLogin();
    else if (optionMain == '2') Registration();
    else if (optionMain == '3') exit(3);
    else {
        cout << "Invalid!";
        system("pause");
        main();
    };
}
```

**Figure 4.2: Function Implementation**

## 4.8 Selection

Selection is a programming technique that allows a program to make decisions based on conditions. The program checks whether a condition is true or false and then runs the appropriate code. Common selection statements include if, if-else, and switch. Selection is used to control the flow of a program, for example choosing which function to execute based on the user's selection.

### 4.8.1 If/Else Technique

```
if (AdminMain == '1')
{
    AdminControlMain(conn);
}
else if (AdminMain == '2')
{
    AdminDisplay(conn, StaffName);
}
else if (AdminMain == '3')
{
    AdminReportMain(conn, StaffName);
}
else if (AdminMain == '4')
{
    main();
}
else {
    cout << "Invalid!" << endl;
    AdminMainMenu(name);
}
```

**Figure 4.3: If/Else Technique Implementation**

### 4.8.2 Switch Technique

```
switch (UpdChoice) {
case '1': column = "StaffName"; break;
case '2': column = "StaffGender"; break;
case '4': column = "StaffPhone"; break;
case '5': column = "StaffEmail"; break;
case '6': column = "StaffPosition"; break;
case '8': column = "StaffRole"; break;
}
```

**Figure 4.4: Switch Technique Implementation**

## 4.9 Control

Control in programming refers to how the order of execution of statements is managed in a program. It determines which parts of the code run first, which run next, and which may be repeated or skipped, based on conditions or user input. Control is achieved using structures such as sequence, selection, and repetition, allowing the program to flow logically and perform tasks in an organized and predictable manner. For example, the while loop below is a control structure that handles repetition by executing the block of code for each record in the result set.

```
while (rs->next())
{
    found = true;
    cout << "\n--------------------\n";
    cout << "Staff ID: " << rs->getString("StaffID") << endl;
    cout << "Name: " << rs->getString("StaffName") << endl;
    cout << "Gender: " << rs->getString("StaffGender") << endl;
    cout << "Age: " << rs->getInt("StaffAge") << endl;
    cout << "Phone: " << rs->getString("StaffPhone") << endl;
    cout << "Email: " << rs->getString("StaffEmail") << endl;
    cout << "Position: " << rs->getString("StaffPosition") << endl;
    cout << "Role: " << rs->getString("StaffRole") << endl;
}
```

**Figure 4.5: Control Implementation**

## 4.10 Error Handling

Error handling in programming is the process of detecting and responding to unexpected or incorrect input, ensuring that the program can continue running safely or give meaningful feedback to the user. It helps prevent the program from crashing and guides users to correct their actions. For example, in a switch statement, the default case can be used to handle invalid choices.

```
default:
    cout << "Invalid choice!\n";
    return;
```

**Figure 4.6: Error Handling Implementation**

## 4.11 Business Rule Implementation

The Blood Bank Management System enforces several core business rules to ensure proper donor eligibility, blood donation tracking and role-based access. These rules are embedded directly into the system's logic and are enforced through condition checks, role validation, and database constraints.

**Role-Based Access Control**

- Only registered users (admin, staff, or donor) are allowed to log in.
- Passwords must match the existing user account before access is granted.
- Passwords are securely stored and never displayed in plain text.
- Each username must be unique across the system.
- Users must reset their password through identity verification before creating a new one.

**Data Validation and Error Handling**

- Input validation is implemented throughout the system (e.g., checking for valid donor IDs, dates, and donation records).
- User credentials are securely checked using masked password input and identity verification for password resets.
- Invalid operations are blocked with descriptive error messages.

**Donation Record and Reward Point Management**

- A donor may only earn reward points for valid and completed blood donations.
- If a donation record is incomplete or invalid, reward points are not awarded.
- Donors cannot redeem points if they have pending or invalid donations and are instructed to complete valid donations first.

## 4.12 Complex Calculation Implementation

Complex calculation implementation in programming refers to the process of performing advanced or multi-step mathematical operations within a program to achieve a specific result. This may involve combining arithmetic operations, using formulas, applying logical or statistical methods, or iterating through large sets of data to compute totals, averages, percentages, or other derived values. Implementing these calculations allows the program to automatically process data accurately and efficiently, reducing human error and saving time. For example, calculating a person's age based on their birth date using the current system date.

```
int calculateAge(int year, int month, int day) {
    time_t t = time(nullptr);
    tm now = {};
    localtime_s(&now, &t);

    int age = now.tm_year + 1900 - year;

    if ((now.tm_mon + 1 < month) ||
        ((now.tm_mon + 1 == month) && (now.tm_mday < day))) {
        age--;
    }

    return age;
}
```

**Figure 4.7: Complex Calculation Implementation**

## 4.13 Report Generation for Analysis Implementation

Report generation in programming is the process of automatically creating organized summaries of data to help users understand, interpret, and analyze information. In the implementation phase, reports are generated from collected data such as records from a database by processing, formatting, and presenting the data in a readable form. Proper report generation helps in decision-making, tracking performance, and identifying trends, making the system more useful and informative for its users. For example, the following while loop processes the donation records retrieved from the database and displays them in a table.

```
while (rs->next()) {
    // Fetch data
    string record = rs->getString(1);
    string donorName = rs->getString(2);
    string date = rs->getString(3);
    string volume = rs->getString(4);
    string type = rs->getString(5);

    // Truncate donor name if too long for display
    string displayName = donorName;
    if (displayName.length() > 25) displayName = displayName.substr(0, 22) + "...";

    // Print formatted line to console
    printf("| %-10s | %-25s | %-12s | %-12s | %-8s |\n",
        record.c_str(),
        displayName.c_str(),
        date.c_str(),
        volume.c_str(),
        type.c_str());

    total++;
}
```

**Figure 4.8: Report Generation for Analysis Implementation**

# 5 CONCLUSION

## 5.6 Constraints

1. **Platform and Technology Constraints**

- The system is developed in C++ with MySQL integration, meaning that portability is limited to machines that support compatible compilers, MySQL connector libraries, and correct configurations.

- The application runs on a text-based user interface. Because of this, user experience is limited and not as intuitive compared to GUI or web-based blood bank systems used in real hospitals.

- The application is designed for single-device local usage. Network support, multi-branch synchronization, and cloud deployment are not currently implemented.

- The system is designed for single-hospital environment. Multi-hospital linking, branch synchronization, and inter-hospital stock sharing is not supported.

2. **Database Constraints**

- The system uses multiple foreign key relationships (Donor → Donation, Donation → BloodStock, etc.). Due to referential integrity rules, some records cannot be directly deleted. For example, deleting a donation that is linked to a blood stock.

- To mitigate this, records may need to be archived instead of physically deleted to maintain traceability in medical data.

- Expiry tracking for blood relies heavily on correct input. There is no automation or trigger-based expiration handling inside the database.

3. **User Role Limitation**

- The system only supports three defined roles: Admin, Staff, and Donor.
- Role permissions are fixed in code. There is no flexible permission policy, privilege control, or support for hospital managers or National Blood Bank central authority roles.
- Authentication is basic; no password hashing, encryption, or MFA is implemented.

4. **Business Logic Constraints**

- Blood volume, blood type compatibility, and transfusion eligibility rules are simplified. Real medical restrictions like RH factor compatibility checking or health risk rejection criteria are not deeply validated.
- Reward points system is simplified and may not reflect real hospital incentive models.
- Emergency urgent request handling is not automated; staff must manually check inventory and availability.

5. **Reporting and Export Constraints**

- Reports are basic: staff report and donation report. Advanced analytics such as blood type demand prediction, donation trend statistics, or cross-branch sharing are not available.

6. **Security Limitations**

- Passwords are masked during input, but they are stored as plaintext in the database, which poses security risks.
- There is no encryption applied to sensitive personal medical data (donor health info, contact details).

## 5.7 Future Improvements

Although the current Blood Bank Management System fulfills its essential operational requirements for managing blood donations, donor information, and reward point tracking within a single hospital environment, several improvements can be implemented in future development to enhance functionality, automation, and scalability.

### 1. Multi-Hospital Support

At present, the system is designed for only one hospital. In the future, the system could be upgraded to support a multi-hospital network environment. This enhancement would allow:

- transferring and sharing blood stock across different hospitals
- centralized blood bank monitoring
- standardized nationwide donor data access

This extension would be suitable for larger organizations, government blood services, and inter-branch communication.

### 2. Automated Eligibility Checking

Currently, eligibility checks (e.g., last donation time, minimum weight, medical condition) still rely on staff evaluation. In the future, the system can implement automatic rule checking, including:

- block donation if required waiting period since last donation date has not passed yet
- restrict donation if health/physical data not within safe range
- automatic notification to donor when next donation date is available again

This improves safety, accuracy, and reduces manual decision errors.

### 3. Mobile Application Integration

Future version could include mobile application support to allow donors to:

- view donation history and points on phone
- receive push reminder when eligible to donate again
- schedule donation appointments online
- receive urgent blood request notification for emergency blood type shortage

This directly increases donor participation and system engagement beyond local PC access.

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